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A FFT-based formulation for efficient mechanical fields computation in isotropic and anisotropic periodic discrete dislocation dynamics

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Abstract

In this paper, we propose a novel full-field approach based on the fast Fourier transform (FFT) technique to compute mechanical fields in periodic discrete dislocation dynamics (DDD) simulations for anisotropic materials: the DDD-FFT approach. By coupling the FFT-based approach to the discrete continuous model, the present approach benefits from the high computational efficiency of the FFT algorithm, while allowing for a discrete representation of dislocation lines. It is demonstrated that the computational time associated with the new DDD-FFT approach is significantly lower than that of current DDD approaches when large number of dislocation segments are involved for isotropic and anisotropic elasticity, respectively. Furthermore, for fine Fourier grids, the treatment of anisotropic elasticity comes at a similar computational cost to that of isotropic simulation. Thus, the proposed approach paves the way towards achieving scale transition from DDD to mesoscale plasticity, especially due to the method's ability to incorporate inhomogeneous elasticity.

Keywords: dislocation dynamics, fast Fourier transforms, anisotropic elasticity

(Some figures may appear in colour only in the online journal)

1. Introduction

Discrete dislocation dynamics (DDD) has proven to be a powerful tool aiding at understanding the collective behavior of large ensembles of dislocations [1–5] at the mesoscale. Such

simulations have successfully been used to investigate plasticity and strain hardening behaviors and processes at small scales. Well-known applications include the study of plasticity in thin films [6–10] and micro-pillars [11–13], the quantification of the strength and formation likelihood of dislocation junctions in materials with different crystal symmetries [14–16], the unraveling of the formation process of triple dislocation junctions [17], the quantification of latent-hardening effects in cubic and hexagonal systems [18–22], and the quantification of the effect on dislocation interaction with irradiation-induced damage on strength [8, 23, 24].

Conceptually, the general procedure in DDD simulation tools is to adopt a discretization scheme (e.g. nodal-based linear or spline approximation, segment-based with fixed sets of segment orientations) and derive thermodynamically consistent sets of equations of motion for each segment in the dislocation ensemble. The DDD procedure includes two main aspects: (i) the quantification of dislocation motion and (ii) the treatment of intersections between dislocation segments. While the latter—pertaining to core reactions—is generally treated via the use of local topological rules ensuring that dissipation is maximized [4], the former requires a complete and accurate knowledge of the stress state throughout the simulation volume as the driving force for dislocation motion depends on it. In current dislocation dynamics models, the stress generally encompasses two contributions, internal and external, which are usually directly superimposed [25]. The internal stress state arises from the elastic stress field interactions induced by the presence of dislocation segments. To date, in the vast majority of nodal DDD simulations, internal stresses are directly calculated by superposing the stress fields induced by individual segments. Since every dislocation segment elastically interacts with all other dislocation segments present in the volume, the computation of nodal interaction forces is a very computationally expensive procedure, which generally constitutes a bottleneck in conventional DDD simulations. Recently, the development of a non-singular theory leading to an analytical stress formulation in isotropic elasticity has permitted the alleviation of the computational cost associated with the calculation of nodal forces induced by pairs of interacting straight dislocation segments [4, 26]. However, such closed-form solutions do not exist for anisotropic elastic materials. As a result, although specific methods for treating anisotropic elasticity have been developed [27–32], the significant increase in computation time associated with anisotropic calculations has precluded large simulations on such materials thus far.

Regarding the determination of the external stress contribution, different approaches are used depending on the type of boundary conditions and on the desired level of accuracy. For instance, if periodic boundary conditions (PBC) are prescribed, a direct mean-field approach can be employed to obtain the external stress contribution from the imposed loading. With this method, the macroscopic mechanical state is assumed to be uniform throughout the entire simulated volume, and therefore prevents the application of traction boundary conditions or the treatment of elastic inhomogeneities. Alternatively, the finite element method (FEM) provides a full-field framework in which the elastic boundary value problem can be solved for more rigorously [25, 33–35]. This approach is also particularly useful to incorporate the treatment of free surfaces via the introduction of image forces. In this case, special treatment for dislocations intersecting the free surfaces must be included [9, 36, 37]. However, the computational cost associated with the FEM prevents the use of fine meshes or limits the achievable level of strain.

As an alternative DDD approach, Lemarchand and co-workers [38] proposed a framework based on an eigenstrain formalism, the discrete continuous model (DCM), in which each dislocation is treated as a plate-like inclusion. By coupling this approach with a FEM framework, the stress state can be directly computed from a classical elasto-viscoplastic constitutive law in which the plastic strain is obtained from the motion of discrete dislocations [39]. In contrast

with conventional DDD approaches relying on the superposition principle, the FEM mesh in the DCM approach accounts for both the boundary conditions and the long-range isotropic or anisotropic elastic interactions. Furthermore, when using PBC, this approach does not require the computation of the elastic stress field of image dislocations [39]. However, while the direct computation of dislocation–dislocation elastic interactions can be significantly reduced with this approach (only the short-range interactions not accounted for by the mesh need to be explicitly computed), the fine FEM meshes required to encompass a substantial amount of long-range interactions and to achieve acceptable accuracy render the approach computationally prohibitive.

In the present study, we propose a new full-field approach based on fast Fourier transforms (FFTs) for the computation of mechanical fields in periodic DDD simulations using the DCM. The use of FFT techniques to solve continuum mechanics problems has, over the past few years, received much attention and has shown to lead to significant computational gains [40]. The FFT-based method for the mechanics of materials was originally developed by Moulinec and Suquet [41, 42] to enable the fast computation of the local responses of composite materials. It was later extended by Lebensohn and co-workers to model the viscoplastic [43] and the elasto-viscoplastic [44] response of polycrystalline aggregates. Regarding dislocation dynamics, FFT-based formulations for computing dislocation elastic interactions have been used in approaches where dislocation lines are not tracked individually, such as in the level set method [45] and in phase field approaches [46, 47]. However, the DD level set method has been reported to be computationally costly due to the need of higher order parametrizations to represent dislocation lines [48], and the formation of junctions, crucial to accurately model strain hardening [4, 17], is not explicitly treated in both approaches. In particular, the absence of the discretization of dislocation lines either restricts junction formations to be purely governed by elastic interactions, generally requiring very small time step increments, or prevents their accurate description. Therefore, in the present work, we propose to implement the FFT-based approach at the length scale of DDD simulations, so as to benefit from the computational efficiency of the FFT method while adopting a discrete representation of dislocations.

The paper is organized as follows: in section 2, the current approaches used in DDD to compute the mechanical fields are presented and their limitations are discussed. Then, the new DDD-FFT method proposed in this paper and its numerical implementation are presented in section 3. Finally, preliminary results on single-crystal DDD simulations are reported in section 4. The interest and validity of the approach are demonstrated by considering four different configurations: (i) a single prismatic dislocation loop in an isotropic medium, (ii) a prismatic loop dipole in an isotropic medium, (iii) a single prismatic loop in an anisotropic medium, and (iv) the activation of a Frank Read loop. Finally, the numerical efficiency of the approach is compared to current methods, while broader perspectives offered by the use of this novel method are discussed in section 5.

2. Current nodal discrete dislocation dynamics approaches

2.1. Dislocation motion

In the nodal approach, dislocation lines are discretized into segments connected to one another through dislocation nodes. The velocity vector $\mathbf{v}_i$ of a dislocation node i is related to the force $\mathbf{F}_i$ acting at the node through the mobility function $\mathcal{M}$:

$$\mathbf{v}_i = \mathcal{M}(\mathbf{F}_i) \quad (1)$$

The mobility function is chosen such that the dislocation motion reproduces well both experimental observations and atomistic simulations. Generally, a linear mobility law is used such that the velocity the dislocation node is directly related to the dislocation driving force as:

$$\mathbf{B}\mathbf{v}_i = \mathbf{F}_i \quad (2)$$

where $\mathbf{B}$ denotes the drag coefficient matrix in the overdamped equation of motion (2). The nodal force $\mathbf{F}_i$ acting at dislocation node i is determined by assembly (in the FEM sense) of segment forces, $\mathbf{F}_i = \sum_j \mathbf{f}_{ij}$, where $\mathbf{f}_{ij}$ denotes the force on dislocation segment ij acting at node i , and where the summation is performed over all nodes j connected to node i . For each dislocation segment ij defined between end nodes $\mathbf{x}_i$ and $\mathbf{x}_j$ and with burgers vector $\mathbf{b}_{ij}$, the force $\mathbf{f}_{ij}$ is obtained by integration of the Peach–Koehler force $\mathbf{f}_{ij}^{pk}(\mathbf{x})$ along its length as:

$$\mathbf{f}_{ij} = \int_{\mathbf{x}_i}^{\mathbf{x}_j} N_i(\mathbf{x}) \mathbf{f}_{ij}^{pk}(\mathbf{x}) d\mathbf{x} \quad (3)$$

with $\mathbf{f}_{ij}^{pk}(\mathbf{x}) = [\boldsymbol{\sigma}(\mathbf{x}) \cdot \mathbf{b}_{ij}] \times \mathbf{t}_{ij}$

where $N_i(\mathbf{x})$ denotes the interpolation function associated with node i , $\mathbf{t}_{ij}$ is the unit tangent of dislocation segment ij , and operator $\times$ is the vector cross product such that $\mathbf{b}_{ij} \times \mathbf{t}_{ij}$ defines the dislocation slip plane. Thus, equations (1)–(3) define the motion of dislocations when subjected to a total stress $\boldsymbol{\sigma}$. As the dislocation lines travel, they shear the crystal, thereby accommodating the imposed deformation. The resulting plastic shear increment $d\gamma^s$, corresponding to the area swept during time dt by the dislocation lines belonging to system s and contained in an elementary volume V , is generally expressed as:

$$d\gamma^s = \frac{b^s \sum_{ij \in s} dA_{ij}}{V} \quad (4)$$

where b^s denotes the magnitude of the Burgers vector on system s . $\sum_{ij \in s} dA_{ij}$ is the sum of areas dA_{ij} sheared by each dislocation segment ij defined between nodes i and j belonging to system s during time dt , and V denotes the sheared volume. After each time increment dt , the nodal displacement $\mathbf{u}_i$ of every dislocation node i can be determined upon time integration as $\mathbf{u}_i = \mathbf{v}_i dt$. The determination of the sheared area dA_{ij} consists in integrating segment ij displacement over its length, and is hence related to the stress $\boldsymbol{\sigma}$ via the Peach–Koehler force. With that, the plastic strain increment $d\boldsymbol{\epsilon}^p$ induced at each time step dt is computed from the collective dislocation motion on all slip systems as:

$$d\boldsymbol{\epsilon}^p = \frac{1}{2} \sum_s (\mathbf{b}^s \otimes \mathbf{n}^s + \mathbf{n}^s \otimes \mathbf{b}^s) d\gamma^s \quad (5)$$

Therefore, as indicated by equations (1)–(5), the determination of dislocation motion in DDD simulations requires the precise evaluation of the total stress $\boldsymbol{\sigma}(\mathbf{x})$ acting along the dislocation lines. In practice, different approaches to determine the mechanical state within the material have been developed and are presented in the next sections.

2.2. Elastic boundary value problem approach

In a vast majority of DDD simulations, the framework used to solve the mechanical boundary value problem generally follows the work of Van der Giessen and Needleman [25]. In the rest

of the paper, DDD simulations adopting this framework, i.e. making use of the superposition principle, will be referred to as regular or conventional approaches. Essentially, the mechanical state at each point $\mathbf{x}$ of the simulation volume V_s must respect the mechanical equilibrium and satisfy the boundary conditions according to the fundamental boundary value problem in mechanics:

$$\begin{aligned}\boldsymbol{\sigma}(\mathbf{x}) &= \mathbb{C} : \boldsymbol{\epsilon}^e(\mathbf{x}), & \forall \mathbf{x} \in V_s \\ \text{div}(\boldsymbol{\sigma}) &= 0 \\ \mathbf{u} &= \mathbf{u}^* \text{ on } \partial V_u \\ \boldsymbol{\sigma} \cdot \mathbf{n} &= \mathbf{t}^* \text{ on } \partial V_t\end{aligned}\tag{6}$$

where $\mathbb{C}$ denotes the fourth order elasticity tensor, $\boldsymbol{\epsilon}^e$ is the elastic strain, and ∂V_u and ∂V_t denote the external surfaces of volume V_s with normal $\mathbf{n}$ subjected to displacement $\mathbf{u}^*$ and traction $\mathbf{t}^*$ boundary conditions, respectively. As mentioned before and introduced in [25], the total stress $\boldsymbol{\sigma}$ can be decomposed as the sum of two contributions: 1) internal stresses $\boldsymbol{\sigma}^{\text{int}}$ arising from the presence of other dislocations, and 2) external stress $\boldsymbol{\sigma}^{\text{ext}}$ coming from the imposed boundary and loading conditions. In most DDD approaches, the body containing dislocations is considered as elastic—outside of the dislocation cores—such that both contributions are directly superimposed to define the total stress $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\text{int}} + \boldsymbol{\sigma}^{\text{ext}}$. Depending on the boundary conditions, the external contribution $\boldsymbol{\sigma}^{\text{ext}}$ is also referred to as the correction field required to ensure that the total stress $\boldsymbol{\sigma}$ respects the boundary conditions and the mechanical equilibrium stated in (6).

2.2.1. Internal stress contribution. In the regular approach, the internal stresses $\boldsymbol{\sigma}^{\text{int}}$ are calculated by virtue of the superposition principle, whereby the contributions of the elastic stress field of each dislocation segment computed in an infinite medium are added. In its general form, the line integral expression of the stress induced by a dislocation loop with Burgers vector $\mathbf{b}$ at any field point $\mathbf{x}$ was obtained by Mura from the eigenstrain theory as [49]:

$$\boldsymbol{\sigma}_{ij}(\mathbf{x}) = \mathbb{C}_{ijkl} \oint_L \epsilon_{lmn} \mathbb{C}_{pqmn} G_{kp,q}(\mathbf{x} - \mathbf{x}') b_m dx'_h\tag{7}$$

where ϵ_{ijk} is the permutation tensor, $G_{ij,k}$ is the derivative of the Green function, and $\mathbf{x}'$ is the coordinate that spans the dislocation line L . This formulation stems directly from the boundary value problem (6) expressed in an infinite medium, i.e. for which no boundary conditions are prescribed, and in which the stress results from the plastic distortion induced by the dislocation line. Following the discretization of the dislocation lines used in DDD, integral (7) is generally decomposed as the sum of integrals over individual connected segments. Furthermore, note that the stress calculated using this formula cannot be determined uniquely when evaluated on a terminating segment [50].

As implicitly expressed in equation (3), the quantification of the force on each dislocation segment requires the evaluation of the stress induced by all other segments present in the simulation. Thus, the calculation of all nodal forces scales with $O(N_{\text{seg}}^2)$ where N_{seg} is the total number of dislocation segments. Following the work of Cai and co-workers [26], a non-singular stress field induced by the presence of a dislocation line in an isotropic medium has been conveniently obtained analytically everywhere in space. This formulation results from introducing a spreading of the Burgers vector so as to remove the singularity in the dislocation core in equation (7). As a result, an analytical expression of the nodal forces pertaining to internal stresses has been derived for elastically isotropic materials, allowing for substantial savings in terms of computation time [4]. However, in the case of anisotropic elasticity, the absence of an

analytical formulation for interaction stresses has thus far prevented any relevant simulations in terms of dislocation densities or achievable strain.

In parallel one has to consider that PBCs are often employed in DDD simulations as they are particularly adapted to studying plasticity in the bulk. When PBCs are prescribed, the unit cell, also referred to as the primary volume, is replicated in all directions and all dislocations exiting a surface are periodically reintroduced through the opposite face, so as to ensure the conservation of dislocation fluxes. In such a case, the stress fields pertaining to the image dislocation segments belonging to the replicated volumes also need to be accounted for in the computation of σ^{int} .

Although approximation procedures for far-field elastic interactions such as the box method [51] or the fast multipole method (FMM) [4, 52] have permitted significant gains in computation time, dislocation–dislocation elastic interactions still remain the most intensive computation procedure in conventional DDD simulations.

2.2.2. External stress contribution. In the literature, the external stress contribution σ^{ext} is often referred to as a correction field required for the total stress σ to satisfy the imposed loading conditions [25, 36]. In practice, the method used for its determination depends on the type of boundary conditions that are prescribed.

The most straightforward approach through which the external stress contribution can be determined relies on the assumption that the boundary conditions generate an homogeneous and uniform stress distribution throughout the whole simulation volume. The obvious advantage of this mean field method lies in the absence of any computational cost since the value of σ^{ext} is readily obtained from the imposed loading through the constitutive law. However, the fact that a single external stress value holds for the entire volume induces important limitations. First, respecting the mechanical equilibrium restricts this method to be only applicable to fully periodic simulations. As a result, the stress field of image dislocations belonging to the replicated volumes must be incorporated and calculated as described in section 2.2.1. Furthermore, this method cannot be employed to treat inhomogeneities.

As an alternative to the mean field direct method, the FEM technique offers a full-field approach in which the boundary value problem can be solved more rigorously. Expressing equation (6) in a weak form and applying the principle of virtual work for any arbitrary virtual displacement δu yields the following expression:

$$\int_{V_s} \sigma_{ij} \delta u_{i,j} dV - \int_{\partial V_s} t_i \delta u_i dS = 0 \quad (8)$$

Examining equation (8), the boundary conditions that can be applied are either (1) nodal displacements and/or (2) stresses (traction conditions) on external surfaces ∂V_s . The second boundary condition type constitutes the primary reason for which the FEM approach has been initially coupled with DDD simulations, thereby allowing for a proper treatment of free surfaces in finite volumes [35, 36]. Recently, the FEM method has also been used and proven successful to model impenetrable interfaces [54] in DDD. As mentioned earlier, in the presence of external or internal surfaces, the FEM approach allows for a consistent numerical treatment of the superposition between (1) the singular problem related to the presence of dislocations in an infinite medium and (2) a correction problem enforcing the true boundary conditions induced by the presence of such surfaces [35, 54]. However, the main limitation of the FEM method for DDD simulations is its relatively intensive computational and memory requirements that prevent the use of very fine meshes, which are often necessary in order to accurately model the mechanical effects, especially in the presence of external or internal surfaces.

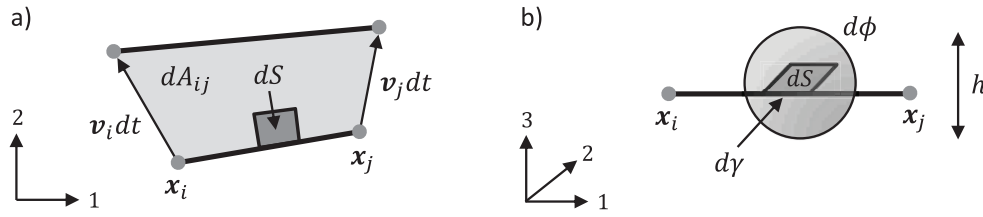


Figure 1. Sheared area dA_{ij} produced by the glide of a dislocation segment ij during time dt . The dislocation segment is defined by its end nodes i and j moving from their initial positions x_i and x_j at velocities v_i and v_j , respectively. Schematic of (a) an elementary sheared area dS swept by the glide of a portion of a dislocation segment, and (b) the resulting elementary homogeneous plastic shear $d\gamma$ associated with an elementary spherical volume $d\phi$ of radius $h/2$ centered on the sheared area.

2.3. The eigenstrain-based approach: the discrete-continuous model

In parallel to the development of the framework used in regular DDD simulations and presented in section 2.2, Lemarchand and co-workers [38] have proposed a different approach to the DDD problem: the DCM. In contrast with the elastic boundary value problem expressed in (6) that relies on the superposition principle, their model is based on an eigenstrain formalism and involves an elasto-viscoplastic framework, such that:

$$\sigma(\mathbf{x}) = \mathbb{C}(\epsilon(\mathbf{x}) - \epsilon^p(\mathbf{x})) \quad (9)$$

in which the plastic strain directly results from the dislocation motion as described by equations (4) and (5). Here, the visco-plastic behavior directly follows from the plastic strain increment $d\epsilon^p$ dependence on the stress (see equations (1)–(5)). Note also that by taking advantage of the time discretization used in DDD, the non-linear constitutive law (9) is often solved via an explicit scheme, whereby the plastic strain state ϵ^p at step n is used to compute the stress and strain at step $n + 1$. Following the eigenstrain formulation, the DCM relies on considering each dislocation loop as a plate-like inclusion of thickness h , such that an elementary sheared area dS swept by the glide of a portion of a dislocation segment ij produces an elementary homogeneous plastic shear $d\gamma$ within an elementary spherical volume $d\phi$ of radius $h/2$ centered on the sheared area dS (see figure 1). One of the most challenging aspects of this approach lies in the regularization of the plastic strain in order to perform accurate and realistic simulations. Specifically, the choice of the elementary sheared volume V in equation (4) is critical since it constitutes a key parameter of the elasto-viscoplastic model. In the recent implementation of Vattré and co-workers [39], the numerical procedure to regularize the plastic shear contribution of a gliding dislocation segment at an integration point of a FEM mesh is as follows: the shear $d\gamma_{ij}^p$ induced by the glide of dislocation segment ij at integration point $\mathbf{p}$ is not null only if $\mathbf{p}$ lies within one or more elementary volume $d\phi(\mathbf{x})$ centered in $\mathbf{x}$ when $\mathbf{x}$ spans the glide area dA_{ij} , i.e.:

$$d\gamma_{ij}^p = \frac{6b}{\pi h^3} \int_{dA_{ij}} \chi(\mathbf{p}, \mathbf{x}) d\mathbf{x} \equiv \frac{6b}{\pi h^3} dS_{ij}^p \quad (10)$$

where b is the magnitude of the segment's Burgers vector, $\pi h^3/6$ is the volume of any elementary sphere $d\phi$ of radius $h/2$, and $\chi(\mathbf{p}, \mathbf{x}) = 1$ if $\mathbf{p}$ lies within the elementary sphere $d\phi(\mathbf{x})$ centered in $\mathbf{x}$, $\chi(\mathbf{p}, \mathbf{x}) = 0$ otherwise. In [39], the numerical integration in equation (10) is performed by successively discretizing the glide area dA_{ij} into elementary areas dS along the segment length and its glide direction.

As discussed in [38], in this approach h is a parameter whose value must be chosen according to the order of the mesh size so as to ensure that the union of the elementary spheres $d\phi$ when dS spans the entire sheared area dA_{ij} at least contains one integration point.

In this approach, the initial mechanical state corresponding to the presence of the initial microstructure has to be taken into account. This is done by introducing initial dislocations according to a Volterra-like process described in [39]. With this procedure, the mechanical fields are fully determined everywhere in the simulation volume, and the resulting stress field encompasses both the external (from the applied loading and the boundary conditions) and internal (from the dislocation microstructure) contributions without redundancies. Specifically, as a result of this regularization procedure, the internal stresses pertaining to long-range dislocation–dislocation elastic interactions are inherently accounted for by the mesh. Here long-range refers to pairs of segments whose interaction distance is larger than half of the regularization parameter h . However, as explained in [39], the finite element mesh does not account for short-range elastic interactions, so that the DCM model requires the computation of interaction forces for pairs of segments which are closer than $h/2$ (see appendix B). Such interactions are added as supplementary local contributions to the Peach–Koehler force.

In the work of Vattré *et al* [39], the DCM method was implemented by coupling it with a FEM solver. However, as mentioned in section 2.2, the expensive computational load associated with the FEM procedure either restricts its use to coarse meshes—therefore requiring the computation of an important number of supplementary local dislocation–dislocation interactions—or makes the procedure impractical due to the computational complexity of 3D FEM. To alleviate such complexity, we propose a novel approach in which the DCM procedure is coupled to a FFT-based solver, aiming at efficiently performing fully periodic DDD simulations in isotropic and anisotropic media.

3. The fast Fourier transform (FFT)-based approach

3.1. FFT-based formulation for DDD

As an alternative full-field stress-strain computational approach, we propose in this work a FFT-based method to be used with the DCM procedure for periodic DDD simulations: henceforth it will be referred to as the DDD-FFT approach. Determining the mechanical state throughout the entire simulation volume V_s in DDD simulations is achieved by solving the fundamental mechanical boundary value problem expressed in equation (6). As described in section 2.3, the constitutive law $\sigma(\mathbf{x}) = \mathbb{C}(\epsilon(\mathbf{x}) - \epsilon^p(\mathbf{x}))$ is linearized by considering the plastic strain $\epsilon^p(\mathbf{x})$, computed using the DCM, as a constant input of the FFT-based solver at each simulation step. Under this approximation and using the small strain hypothesis, the local mechanical equilibrium $\sigma_{ij,j}(\mathbf{x}) = 0$ reads:

$$\begin{aligned} \mathbb{C}_{ijkl} u_{k,lj}(\mathbf{x}) - \varphi_{ij,j}(\mathbf{x}) &= 0, \quad \forall \mathbf{x} \in V_s \\ \text{where } \varphi_{ij}(\mathbf{x}) &= \mathbb{C}_{ijkl} \epsilon_{kl}^p(\mathbf{x}) \end{aligned} \quad (11)$$

Assuming the mechanical fields are periodic in the three spatial directions and denoting ξ the (frequency) coordinate on the Fourier space, the equilibrium equation (11) in the Fourier space writes:

$$\mathbb{C}_{ijkl} \xi_l \xi_j \hat{u}_k(\xi) + \hat{\varphi}_{ij,j}(\xi) = 0 \quad (12)$$

where $\hat{\mathbf{u}}$ and $\hat{\boldsymbol{\varphi}}$ denote the Fourier transforms of the displacement field $\mathbf{u}$ and the polarization tensor $\boldsymbol{\varphi}$ is defined in the real space, respectively. If we denote $K_{ik} = \mathbb{C}_{ijkl}\xi_l\xi_j$, the displacements in the Fourier space can be obtained from equation (12) as:

$$\begin{aligned} \hat{u}_i(\boldsymbol{\xi}) &= -\hat{G}_{ik}(\boldsymbol{\xi})\hat{\varphi}_{kj,j}(\boldsymbol{\xi}), \quad \forall \boldsymbol{\xi} \neq 0 \\ \text{where } \hat{G}_{ik}(\boldsymbol{\xi}) &= K_{ki}^{-1} = [\mathbb{C}_{kjil}\xi_l\xi_j]^{-1} \end{aligned} \quad (13)$$

In equation (13), $\hat{G}(\boldsymbol{\xi})$ denotes the periodic Green's function expressed in the Fourier space. Taking advantage of the spatial derivative property of the Fourier transform, relation (13) can be conveniently rewritten as:

$$\hat{u}_i(\boldsymbol{\xi}) = -i\xi_l \hat{G}_{ik}(\boldsymbol{\xi})\hat{\varphi}_{kl}(\boldsymbol{\xi}), \quad \forall \boldsymbol{\xi} \neq 0 \quad (14)$$

where $i = \sqrt{-1}$ denotes the imaginary number in the complex space. With equation (14), the strain field $\hat{\boldsymbol{\varepsilon}}(\boldsymbol{\xi})$ can be directly calculated in the Fourier space as:

$$\begin{aligned} \hat{\varepsilon}_{ij}(\boldsymbol{\xi}) &= \frac{1}{2}(\hat{u}_{i,j}(\boldsymbol{\xi}) + \hat{u}_{j,i}(\boldsymbol{\xi})) = \hat{\Gamma}_{ijkl}(\boldsymbol{\xi})\hat{\varphi}_{kl}(\boldsymbol{\xi}), \quad \forall \boldsymbol{\xi} \neq 0, \quad \hat{\varepsilon}_{ij}(0) = 0 \\ \text{where } \hat{\Gamma}_{ijkl}(\boldsymbol{\xi}) &= \frac{1}{2}(\xi_l\xi_j\hat{G}_{ik}(\boldsymbol{\xi}) + \xi_l\xi_i\hat{G}_{jk}(\boldsymbol{\xi})) \end{aligned} \quad (15)$$

where $\hat{\Gamma}(\boldsymbol{\xi})$ is the modified Green's function. For DDD-FFT simulations in which the primary volume is discretized into a regular 3D grid of N_i Fourier (grid) points in each direction $i = 1, 3$, the polarization tensor $\boldsymbol{\varphi}(\mathbf{x})$ related to the plastic strain via equation (11) can be determined at every grid point $\mathbf{x}$ in the real space through the DCM procedure as described in section 2.3. Consequently, its discrete Fourier transform $\hat{\boldsymbol{\varphi}}(\boldsymbol{\xi})$ at corresponding frequencies $\boldsymbol{\xi}$ required in the calculation of $\hat{\boldsymbol{\varepsilon}}(\boldsymbol{\xi})$ in equation (15) can be easily computed via the use of the FFT algorithm. Similarly, the FFT algorithm can be used to compute the inverse discrete Fourier transform of $\hat{\boldsymbol{\varepsilon}}(\boldsymbol{\xi})$. Note that the solution obtained from taking the inverse Fourier transform of equation (15) gives the fluctuations from the mean strain $\langle \boldsymbol{\varepsilon} \rangle$, where $\langle \cdot \rangle$ denotes the spatial average over the primary volume V_s . Therefore, by imposing the condition $\hat{\boldsymbol{\varepsilon}}(0) = 0$ on the average in equation (15), the corresponding strain field $\boldsymbol{\varepsilon}(\mathbf{x})$ in the real space is obtained by:

$$\boldsymbol{\varepsilon}_{ij}(\mathbf{x}) = \mathcal{FFT}^{-1}(\hat{\boldsymbol{\varepsilon}}_{ij}(\boldsymbol{\xi})) + E_{ij} \quad (16)$$

where $\mathcal{FFT}^{-1}$ denotes the inverse discrete Fourier transform operation and $\mathbf{E} = \langle \boldsymbol{\varepsilon} \rangle$ corresponds to the average imposed macroscopic strain. With this approach, the primary volume can be seen as being periodically replicated in all spatial directions and immersed in the material bulk subjected to a macroscopic strain $\mathbf{E}$. From equation (16), the stress field at each grid point $\mathbf{x}$ is directly obtained from the compatible strain using the constitutive law:

$$\boldsymbol{\sigma}(\mathbf{x}) = \mathbb{C}(\boldsymbol{\varepsilon}(\mathbf{x}) - \boldsymbol{\varepsilon}^p(\mathbf{x})) \quad (17)$$

With this formulation, the stress field $\boldsymbol{\sigma}(\mathbf{x}) = \boldsymbol{\sigma}^{\text{int}}(\mathbf{x}) + \boldsymbol{\sigma}^{\text{ext}}(\mathbf{x})$ encompasses both the contributions of the dislocations elastic stress field and the stresses arising from the imposed loading conditions without redundancy. Furthermore, by performing a simple interpolation between the grid points, the total stress state can be obtained everywhere in the simulation volume, particularly along every dislocation segment such that the Peach–Köhler force in equation (3) can be determined. However, as mentioned earlier, the grid does not account for local interactions when using the DCM approach. As a result, a local contribution must be added to

equation (17) when computing the Peach–Koehler force along dislocation segments whose neighbors are closer than $h/2$. Full details about this procedure are presented in appendix B.

In addition, note that for any arbitrary dislocations distribution and when no loading is applied, i.e. for $\mathbf{E} = 0$, equations (16) and (17) yield the internal stress field $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\text{int}}$ solely pertaining to the presence of the dislocations in the load-free infinite periodic medium with elastic stiffness $\mathbb{C}$. However, outside of the dislocation cores, this stress would not necessarily coincide with the analytical stress $\tilde{\boldsymbol{\sigma}}^{\text{int}}$ computed for the same microstructure using Mura line integral formula (7) or the non-singular formulation in [26]. In fact, as detailed in appendix A, both evaluations are generally obtained up to a constant translation tensor $\bar{\boldsymbol{\sigma}}$ such that $\boldsymbol{\sigma}^{\text{int}}(\mathbf{x}) = \tilde{\boldsymbol{\sigma}}^{\text{int}}(\mathbf{x}) + \bar{\boldsymbol{\sigma}}$, $\forall \mathbf{x} \in V_s$. This is because the eigenstrain boundary value problem is not solved for the same boundary conditions when using the FFT-based approach or using the Mura formula [49]. In the case where local interactions are to be added to equation (17), the analytical stress formulations are used, such that $\bar{\boldsymbol{\sigma}}$ needs to be determined. More details on this specific aspect and on a method to compute $\bar{\boldsymbol{\sigma}}$ are provided in appendices A and B.

Besides, note that the simulation can be alternatively driven in stress by substituting the imposed value of the macroscopic strain $\mathbf{E}$ in equation (16) with its resultant when imposing a macroscopic stress $\boldsymbol{\Sigma}$ as:

$$\mathbf{E}_{ij} = [\mathbb{C}_{ijkl}]^{-1} \boldsymbol{\Sigma}_{kl} + \boldsymbol{\epsilon}^p_{ij} \quad (18)$$

3.2. Numerical implementation

The FFT-based solver and the DCM procedure have been implemented in the DDD code employed in [22]. While the implementation of the FFT-based solver does not present any particular difficulty (the numerical difficulties associated with it will be treated in section 4), the DCM procedure requires a specific implementation detailed in this section.

Following the DCM method described in section 2.3, the plastic shear $d\gamma_{ij}^p$ produced by the glide of a dislocation segment ij can be regularized at each grid point $\mathbf{p}$ according to equation (10) as:

$$d\gamma_{ij}^p = \frac{b \, dS_{ij}^p}{V} \quad (19)$$

where $V = \pi h^3/6$ denotes the elementary sphere volume of radius $h/2$ in equation (10). In [39], the integration in equation (10) to determine dS_{ij}^p is numerically performed by the discretization of the glide area dA_{ij} into elementary areas dS . However, this formulation is limited to the case of trapezoidal glide areas associated with segment-based discretization approaches in which dislocation segments do not rotate in their slip plane.

In this work, we propose an alternative analytical method to perform the integration in equation (10) that should be suitable for general dislocation motions in nodal-based discretization schemes, i.e. compatible with the rotation of dislocation segments in their slip plane. First, one can notice that computing dS_{ij}^p is equivalent to computing the intersection between the sphere $d\phi(\mathbf{p})$ of radius $h/2$ centered at integration point $\mathbf{p}$ and the quadrilateral defined by the segment sheared area defined in figure 1. In our formulation, $\mathbf{p}$ now refers to the position of any point of the Fourier grid. Based on the fact that the intersection between a sphere and a quadrilateral reduces to an in-plane intersection between a circle (resulting from the intersection between the sphere and the plane containing the quadrilateral) and a quadrilateral, the contour of such an intersection is therefore exclusively composed of a succession of straight

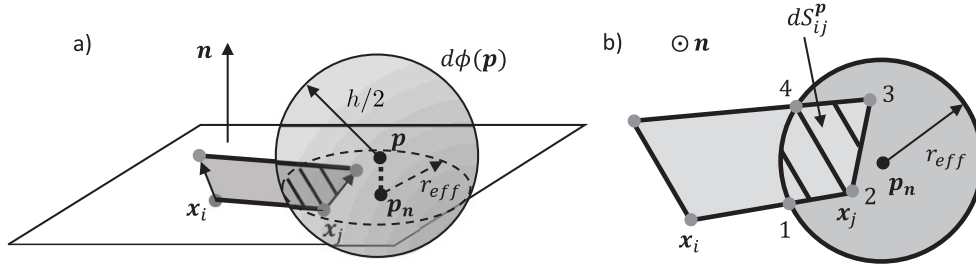


Figure 2. (a) Example of the intersection between the area swept by a dislocation segment ij gliding on a plane with unit normal $\mathbf{n}$ and the elementary sphere $d\phi(\mathbf{p})$ of radius $h/2$ centered in grid point $\mathbf{p}$. The intersection between the sphere $d\phi(\mathbf{p})$ and the glide plane of the dislocation results in a circle of radius $r_{\text{eff}} = \sqrt{h^2/4 - d^2}$ and of center $\mathbf{p}_n$ where $\mathbf{p}_n$ is the orthogonal projection of $\mathbf{p}$ onto the dislocation plane such that $\mathbf{p} - \mathbf{p}_n = d\mathbf{n}$. (b) The intersection area dS_{ij}^p (shaded region) can be analytically calculated using Green's theorem by following the oriented contour composed of the straight segments $\overline{12}$, $\overline{23}$ and $\overline{34}$ and the arc 41 .

segments and arcs, as depicted in figure 2. Following that, dS_{ij}^p can be analytically computed by line integration using Green's theorem:

$$dS_{ij}^p = \frac{1}{2} \oint_{C_{ij}^p} (-y \, dx + x \, dy) \quad (20)$$

where C_{ij}^p denotes the closed contour of the intersection between the quadrilateral defined by the motion of segment ij and the sphere $d\phi(\mathbf{p})$ centered in grid point $\mathbf{p}$, and x and y are the coordinates spanning the contour in the 2D frame defined in the dislocation glide plane. For a closed contour formed of n successive straight segments and arcs, expression (20) can be further decomposed as the summation of individual line integrals (see figure 2(b)):

$$dS_{ij}^p = \sum_{k=1}^n \frac{1}{2} \int_{C_k} (-y \, dx + x \, dy) = \sum_{k=1}^n I_{C_k} \quad (21)$$

where $\{C_k, k = 1, \dots, n\}$ denotes the piecewise continuous set of individual curves (either straight segments or arcs) defining the full contour C_{ij}^p . For any straight segment C_k defined between vertices (x_0, y_0) and (x_1, y_1) , the line integral $I_{C_k}^{\text{seg}}$ along this segment can be analytically derived from equation (20) as:

$$I_{C_k}^{\text{seg}} = \frac{1}{2} (x_0 y_1 - y_0 x_1) \quad (22)$$

Similarly, for any arc C_k defined as a portion of a circle of radius r centered in (x_c, y_c) and delimited by end vertices (x_0, y_0) at angle θ_0 and (x_1, y_1) at angle θ_1 (assuming $\theta_1 > \theta_0$), the line integral $I_{C_k}^{\text{arc}}$ along this arc can be analytically derived from equation (20) as:

$$I_{C_k}^{\text{arc}} = \frac{1}{2} [r^2(\theta_1 - \theta_0) + x_c(y_1 - y_0) - y_c(x_1 - x_0)] \quad (23)$$

The full contour bounding the region dS_{ij}^p is thus determined by ordering and joining the quadrilateral vertices and the circle-quadrilateral intersection points through straight segments and arcs (see figure 2(b)). Note that such a contour can be directly obtained by using

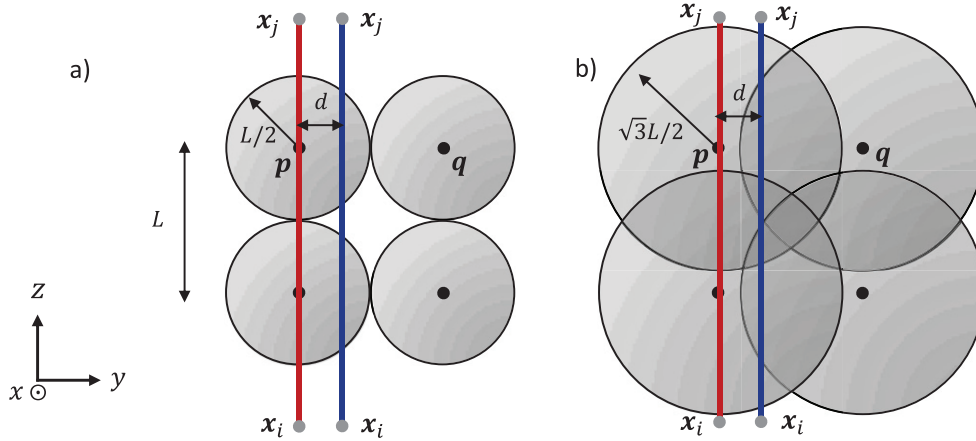


Figure 3. Slice in the (yz) plane of a dislocation segment ij shearing the volume in the (xz) plane. The red dislocation segment is aligned with grid point p while the blue segment is away from a distance d . (a) For $h = L$ the union of all the elementary spheres centered in grid points does not map the entire volume. As a result, the plastic shear induced by the blue dislocation segment ij is not entirely transferred to the mesh, leading to incorrect results. (b) For $h = \sqrt{3}L$ the union of all the elementary spheres maps the entire volume, such that the entire plastic strain is transferred to the mesh. However, the overlapping between the elementary spheres results in a smearing out of the plastic strain that needs to be corrected for.

computational geometry libraries such as CGAL [55]. In the present work, the numerical algorithm detailed in appendix C is implemented.

The choice of the parameter h , corresponding to the diameter of the elementary sphere $d\phi$ associated with each grid point, is critical in this approach. In the current implementation, its value must be chosen such that the union of the spheres associated with every grid point at least maps the entire simulation volume. In other words, care must be taken to ensure that the entire sheared area produced by any dislocation line intersects with elementary spheres, such that the total plastic strain produced is entirely transferred to the mesh. With that, the minimum acceptable value for any regular 3D mesh of size L (i.e. for which the distance between two consecutive grid points is L in each direction) is $h = \sqrt{3}L$ (see figure 3). Note that for this value of h , an overlap between the elementary spheres exists, resulting in a smearing out of the plastic strain, i.e. in a loss of accuracy. Whereas in [38], the value $h = 3/2L$ has been determined to be optimal and the same value has been used in [39], the analytical regularization procedure introduced in the present work induces a dependency of dS_{ij}^p on the position of the dislocation line with respect to the grid (see section 4.1). If the dislocation core is perfectly aligned with grid points (the red dislocation line in figure 3), the obtained stress field is similar to the analytical solution; however inaccuracies occur when the dislocation core lies in between two grid points (the blue dislocation line in figure 3). For this reason, in the following, we propose a method to correct the plastic shear distribution depending on the position of the dislocation with respect to the grid.

According to the description of the regularization procedure, the position dependency of the regularized plastic strain may originate from two sources: (1) the overlap between contiguous elementary spheres, and (2) the evolution of the regularized plastic strain as a function of the distance between the grid points and the dislocation core. To circumvent the first source

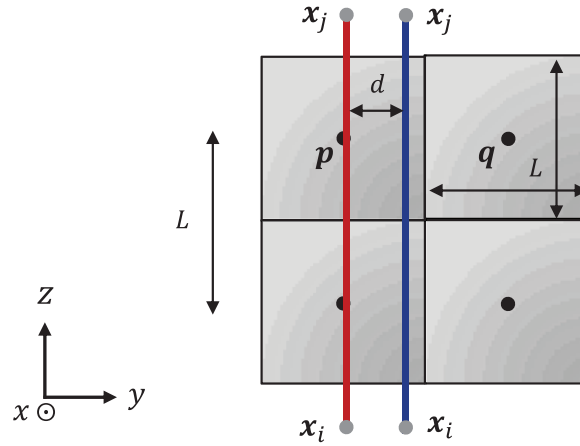


Figure 4. Schematic of two dislocation segments shearing the volume along the (xz) plane. The red dislocation segment is aligned with grid point p while the blue segment is positioned at a distance d from grid point p , hence at distance $L - d$ from the subsequent grid point q . When the elementary volumes associated with each grid point are chosen to be square boxes of side $a = L$, the simulation volume is entirely mapped and no overlapping is present. However both the red and blue dislocation segments would induce a similar plastic shear $d\gamma_{ij}^p = b/L$ (see equation (24)) at grid point p , so that the spatial positioning of the dislocation core is not accounted for.

of position-dependency, let us use an elementary volume such that the union over all the grid points precisely maps the simulation volume, without inducing any overlapping. Only an elementary volume chosen as a square box of side $a = L$ would satisfy such a mapping (see figure 4). With a square box, the plastic strain $d\gamma_{ij}^p$ induced by a dislocation segment ij entirely shearing the elementary box (centered at grid point p) along the (xz) plane would be:

$$d\gamma_{ij}^p = \frac{b L^2}{L^3} = \frac{b}{L} = d\gamma_{ij}^{\text{ref}} \quad (24)$$

This result is simply obtained from equation (19) with $dS_{ij}^p = L^2$ and $V = L^3$ for a dislocation segment entirely shearing a square box of side length L . However, as depicted in figure 4, with such an elementary volume, a red dislocation segment aligned with a grid point p will yield the same plastic strain $d\gamma_{ij}^p$ at point p as a blue dislocation segment positioned at a distance d from the grid point, since the intersection area $dS_{ij}^p = L^2$ is the same in both cases. In other words, using elementary square boxes leads to disregarding the spatial positioning of the dislocation core for any dislocation lying at distance $\pm d$ from a grid point. This issue is precisely related to the second source of position dependency, namely the evolution of $d\gamma_{ij}^p$ with respect to the distance d from the grid point p . Note that for the rest of this section, $d\gamma_{ij}^{\text{ref}}$ as defined in equation (24) would denote the reference amount of regularized plastic shear required to obtain an exact stress field for any red dislocation aligned with the grid, i.e. for which $d = 0$.

However, a linear interpolation of the plastic shear with respect to the distance d between the dislocation plane and the grid point ($0 < d < L$) is seen to yield correct stress field values (this will be later shown in figure 8). Such a linear interpolation between two successive grid points can be effectively obtained by using an elementary spherical volume and following the procedure detailed hereafter (the sphere is chosen such that the method remains valid

regardless of the orientation of the dislocation glide plane). Using expression (19) and considering a sphere of radius r fully sheared, the evolution of $d\gamma_{ij}^p(d)$ with respect to the distance d to grid point $\mathbf{p}$ can be expressed as:

$$d\gamma_{ij}^p(d) = \frac{b \pi r_{\text{eff}}^2}{(4/3)\pi r^3} = \frac{b (r^2 - d^2)}{(4/3)r^3} \quad (25)$$

where $r_{\text{eff}} = \sqrt{r^2 - d^2}$ denotes the effective radius of the circle resulting from the intersection between the elementary sphere $d\phi(\mathbf{p})$ and the glide plane of the dislocation, as depicted in figure 2. Clearly, when using a sphere as the elementary volume, $d\gamma_{ij}^p(d)$ is not a linear function of d . However, it can be corrected such that $d\gamma_{ij}^{p,\text{corr}}(d)$ becomes a linear function of d . To obtain a linear interpolation, such a corrected function must be bounded by $d\gamma_{ij}^{p,\text{corr}}(0) = d\gamma_{ij}^{\text{ref}}$ and $d\gamma_{ij}^{p,\text{corr}}(L) = 0$. In order to fulfill these requirements, the sphere radius can be conveniently chosen as $r = L$, i.e. $h = 2r = 2L$. With that, expression (25) is written:

$$d\gamma_{ij}^p(d) = \frac{b (L^2 - d^2)}{(4/3)L^3} \quad (26)$$

The linear interpolation with respect to d is obtained for $0 < d < L$ if $d\gamma_{ij}^{p,\text{corr}}(d) = (L - d)/L \cdot d\gamma_{ij}^{\text{ref}}$. Using equation (26), and denoting $c(d)$ the correction function defined such that $d\gamma_{ij}^{p,\text{corr}}(d) = c(d) \cdot d\gamma_{ij}^p(d)$, one can write:

$$d\gamma_{ij}^{p,\text{corr}}(d) = c(d) \cdot d\gamma_{ij}^p(d) = \frac{(L - d)}{L} d\gamma_{ij}^{\text{ref}} \quad (27)$$

Using the definitions of $d\gamma_{ij}^{\text{ref}}$ and $d\gamma_{ij}^p(d)$ from equations (24) and (26), the correction function $c(d)$ can be determined from equation (27) as:

$$c(d) = \frac{4 (L^2 - Ld)}{3 (L^2 - d^2)} \quad (28)$$

Essentially, equation (27) states that for $h = 2L$, the plastic strain increment $d\gamma_{ij}^p$ produced by any dislocation shearing the spherical elementary volume $d\phi(\mathbf{p})$ needs to be corrected by a position dependent factor $c(d)$, whose expression is given in equation (28). With such a correction in the regularization procedure, the DDD-FFT approach is found to provide an exact match with the analytical solution for the stress field of dislocation segments, irrespective of their position and orientation on the Fourier grid, as attested in section 4. Therefore, in this work, the diameter of the elementary spheres associated with every grid point is set to $h = 2L$ and the regularized plastic strain computed with expressions (19) and (21) is systematically corrected with equations (27) and (28).

4. Application of the DDD-FFT approach

4.1. Static dislocations in isotropic elasticity

As a first validation of the DDD-FFT approach, the stress field obtained for a prismatic loop is assessed. For the basis of comparison, the set-up of the simulation is similar to that taken in [39]. The simulation cell is assumed to be fully periodic, i.e. infinitely replicated in all spatial directions. As depicted in figure 5(a), the prismatic square loop is composed of four edge segments of length $l_0 = 1000b$ with Burgers vector $\mathbf{b} = [010]$ such that $|\mathbf{b}| = b = 0.25$ nm. The four segments are lying on the (010) plane of a simple cubic lattice, and the loop is first

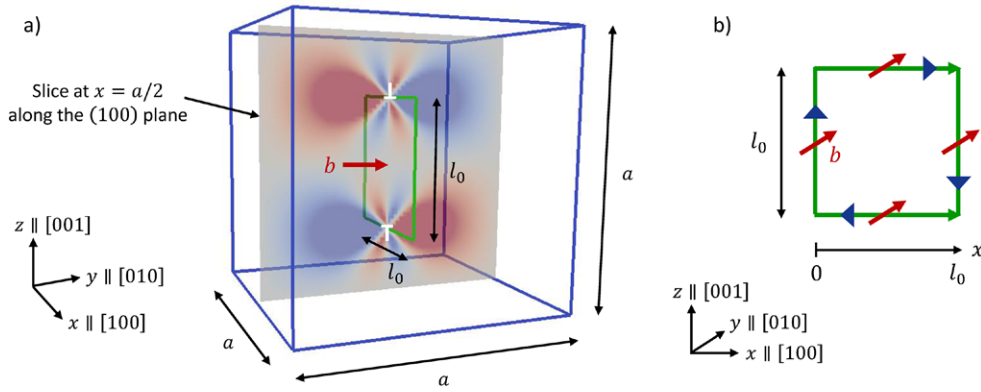


Figure 5. Set-up used for evaluating the stress field of a prismatic loop using the DDD-FFT approach. Only the primary volume of the periodic simulation is shown here. (a) The prismatic dislocation loop is composed of four edge segments of length $l_0 = 1000b$ with Burgers vector $\mathbf{b} = [010]$ and lying on the (010) plane. The loop is initially positioned at the center of the simulation box of side length $a = 0.5 \mu\text{m}$. (b) The prismatic loop is introduced in the simulation following the Volterra-like process: two initial vertical edge segments with the same Burgers vector $\mathbf{b} = [010]$ but opposite line directions are introduced at position $x = 0$. One of the segments is then held at its original position while the second segment is moved apart at a distance l_0 on the (010) plane, thereby creating the closed prismatic loop. The blue arrows indicate the line direction of each segment.

positioned at the center of the simulation box. In order for the mesh size to be identical in each direction, the primary simulation volume V_s is taken as a cube with side $a = 0.5 \mu\text{m}$ such that $V_s = 0.5 \times 0.5 \times 0.5 \mu\text{m}^3$, and discretized into N grid points in each direction, i.e. for a total of $N_t = N \times N \times N$ grid points (or voxels).

The prismatic dislocation loop is introduced in the simulation box following the Volterra-like process: two parallel edge segments of the same Burgers vector but opposite line directions are initially introduced at the same position and are then moved apart along the (010) plane until forming the square prismatic loop (see figure 5(b)). The material is assumed to be elastically isotropic with shear modulus $\mu = 51 \text{ GPa}$ and Poisson's ratio $\nu = 0.37$. The resulting stress field obtained for the prismatic dislocation loop through this process is compared to the analytical solution obtained with the non-singular formulation of Cai and co-workers [26]. Figure 6 shows the σ_{23} component of the resulting stress fields on a (100) slice taken at the center of loop ($x = a/2$) for a Fourier grid made of $64 \times 64 \times 64$ voxels. The oscillations observed in figure 6(b) pertain to the well-known Gibbs phenomenon associated with spectral methods: this phenomenon occurs in taking Fourier transforms of discontinuous fields. Because of the discontinuities in the plastic strain field induced by dislocations, the undesirable Gibbs effect is to be expected. In order to tackle this effect, a recent approach based on a 'numerical spreading' was proposed in the work of Brenner *et al* [56]. In their 2D implementation of static field dislocation mechanics, the authors proposed to spread any dislocation density defined at a pixel across its neighboring pixels and studied the effect of different distribution schemes. It was found that spreading the core of a dislocation over 3×3 pixels using a triangular distribution was sufficient to strongly attenuate the oscillations produced from the Gibbs phenomenon. Consequently, in the present work, a similar 'numerical spreading' of the plastic strain with a 3D triangular distribution over $3 \times 3 \times 3$ voxels is used. Practically, the plastic strain computed at each grid point from equation (27) is distributed over the 27

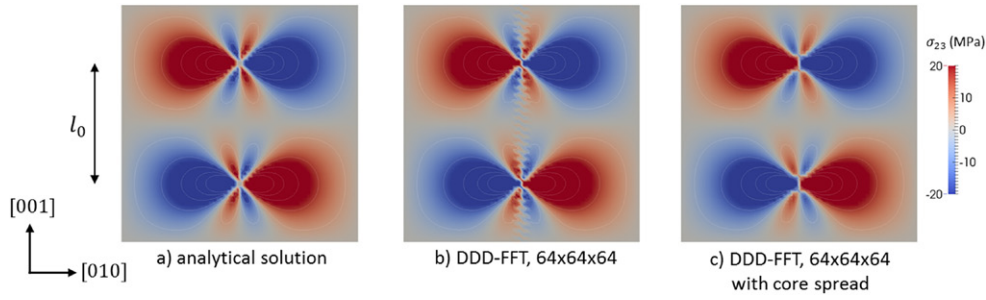


Figure 6. σ_{23} shear component of the stress field of a prismatic loop along a (100) slice taken at $x = a/2$ (see figure 5(a)) obtained: (a) using the analytical solution, (b) using the DDD-FFT approach with a grid of $64 \times 64 \times 64$ voxels. Numerical oscillations pertaining to the Gibbs phenomenon are occurring, and (c) using the DDD-FFT approach with a grid of $64 \times 64 \times 64$ voxels and by spreading the regularized plastic strain over $3 \times 3 \times 3$ voxels. Numerical oscillations are removed and the dislocation core spreads as a result.

neighboring voxels using a triangular distribution. This numerical spreading is somehow analogous to the isotropic core spread introduced in [26], except that the core width is related to the mesh size in this case. Note that other approaches such as to use discrete derivatives in the Fourier space can be alternatively employed to attenuate the Gibbs oscillations [57, 58]. The stress field obtained with the spreading is shown in figure 6(c). One can observe that the oscillations have been removed and that the dislocation core is seen to be spread as a consequence. For the sake of comparison, the stress component σ_{23} is plotted along a line passing through the core of the dislocation, as shown in figure 7(a). While the stress fields obtained with the DDD-FFT approach perfectly match the analytical solution away from the core, the numerical spreading naturally induces a smearing out in the vicinity of the core. Nevertheless, the oscillations as shown in figure 6(b) are not acceptable in DDD since the stress constitutes the driving force of dislocation segments, and must be eliminated. In other words, if such numerical oscillations were present in the resulting stress field, they would inevitably propagate throughout the motion of dislocations, thus leading to the collapse of the whole simulation. For this reason, for the rest of the paper, the presented results will necessarily involve the numerical spreading of dislocation cores.

In line with the examination of the numerical accuracy of the method, the effect of the grid resolution is plotted in figure 7(b) for grids ranging from $16 \times 16 \times 16$ to $128 \times 128 \times 128$ voxels. As expected, a finer grid produces more accurate results as we approach the dislocation core. In addition, a finer mesh will induce a smaller spatial spread of the dislocation core. Besides, it must be recalled that from the DCM procedure, the stress field in the vicinity $h/2$ of the core of a dislocation cannot be accounted for by the mesh [39]. Consequently, for pair of interacting dislocation segments whose stress integration points are closer than $h/2 = L$ (where L is the mesh size), an additional local stress contribution must be added to the total stress interpolated from the Fourier grid as presented in section 3.1. This supplementary contribution is crucial for a proper treatment of junction formation. In the case of isotropic elasticity, the analytical formulation proposed in [4, 26] can be directly used for the calculation of local interaction stresses and forces (see appendix B). In the case of anisotropic calculations for which no analytical solution can be obtained, more expensive methods must be employed [28, 30–32]. Note, however, that in the original DCM procedure, the additional intensive computational layer associated with the FEM solver is restraining its use to relatively coarse meshes,

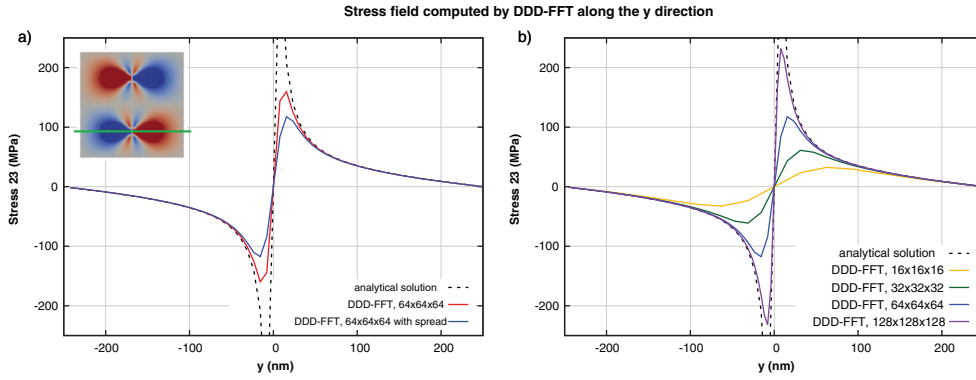


Figure 7. Evolution of the σ_{23} shear component of the stress field of a prismatic loop along the green line shown in the inset. In this frame, the origin of the y-axis is aligned with the position of the dislocation core. (a) Comparison between the analytical solution and the DDD-FFT approach with and without ‘numerical spreading’ for a grid of 64×64 voxels. (b) Comparison between the analytical solution and the result obtained with the DDD-FFT approach including the numerical spreading for different grid sizes.

thus requiring the computation of a substantial number of local interactions. Conversely, the efficiency of the FFT-based approach allows for the use of finer grids, thus significantly reducing the number of local interaction calculations. Therefore, in the case where the mesh size is chosen to be sufficiently small such that a marginal number of local interactions have to be computed, the treatment of anisotropic elasticity comes at a roughly equivalent cost as for isotropic elasticity.

As detailed in section 3.2, the transfer of the plastic shear resulting from dislocation motion to the grid points needs to be corrected using equations (27) and (28). To validate this correction, the stress field of the prismatic loop positioned at three different locations with respect to the grid is compared to the analytical solution. The three positions chosen for the dislocation loop lying on the (010) plane are depicted in figure 8(a) and the resulting stress fields obtained with and without the correction are reported in figure 8(b). As attested by the dashed lines, it clearly appears that the resulting stress field deviates from the analytical solution depending on the position of dislocation with respect to the grid when the correction introduced in equations (27) and (28) is not accounted for. Conversely, the resulting stress fields superimpose irrespective of the position in the grid and converge towards the analytical solution when the plastic shear transfer is corrected, as shown by the solid lines.

4.1.1. Prismatic dipole. To assess the validity of the DDD-FFT approach when multiple dislocations are present in the simulation, the stress field produced by a dislocation dipole is examined here. The prismatic dislocation dipole is created by simultaneously introducing two prismatic loops of opposite Burgers vectors. To ensure that the procedure will be valid in a more general context, the two prismatic loops forming the dipole are positioned in between grid points, respectively, at distances $d = L/4$ and $d = 3L/4$ according to the frame depicted in figure 9(a). The resulting stress field shown in figure 9(b) is seen to perfectly match the analytical solution given by [26] away from the direct vicinity of the dislocation cores. This example assesses the ability of the FFT-based technique to achieve direct superposition of the mechanical fields of distinct dislocations, hence validating the DDD-FFT approach to simultaneously handle multiple dislocations.

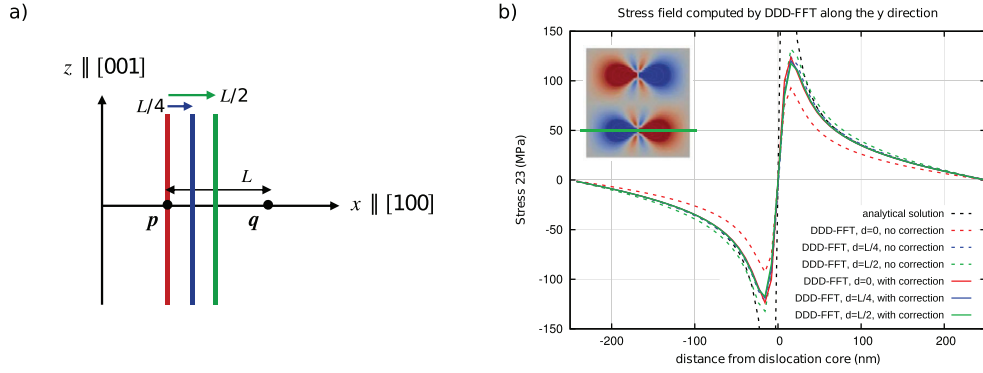


Figure 8. (a) Schematic of the different tested positions for the (010) prismatic dislocation loop with respect to the Fourier grid. Two consecutive grid points p and q are separated by the mesh size distance L and the red, blue, and green dislocation segments introduced as depicted in figure 5 are positioned at distances $d = 0, \frac{L}{4}, \frac{L}{2}$ from grid point p , respectively. (b) The effect of the correction on shear component σ_{23} of the stress field of a prismatic loop along the green line shown in the inset as a function of the dislocation position with respect to the Fourier grid made of $64 \times 64 \times 64$ voxels.

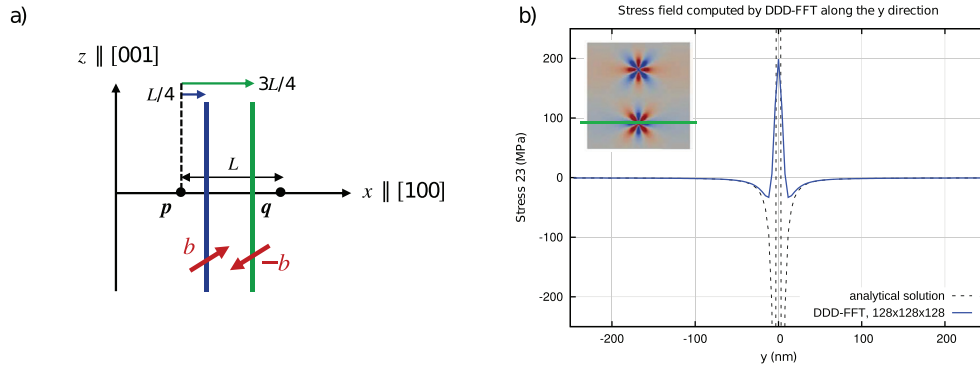


Figure 9. (a) Schematic of the positioning of the two prismatic dislocation loops of the opposite Burgers vectors $b = [010]$ forming the dipole. (b) Comparison of the evolution of the σ_{23} shear component of the stress field of a prismatic dipole along the green line shown in the inset between the analytical solution and the result obtained with the DDD-FFT approach for a resolution of $128 \times 128 \times 128$ voxels. In this frame, the origin of the y -axis is taken at the middle of the two dislocation loops.

4.2. Anisotropic elasticity

As detailed in section 3.1, solving the mechanical equilibrium in the Fourier space enables the direct use of the periodic Green's function through its definition, expressed in equation (13). Therefore, computing the exact stress state—outside of the direct vicinity of the core—associated with periodic dislocations in an anisotropic medium comes at no supplementary cost with respect to an isotropic medium. In this section the aim is to show the applicability of the DDD-FFT method developed in this work to the case of anisotropic elasticity.

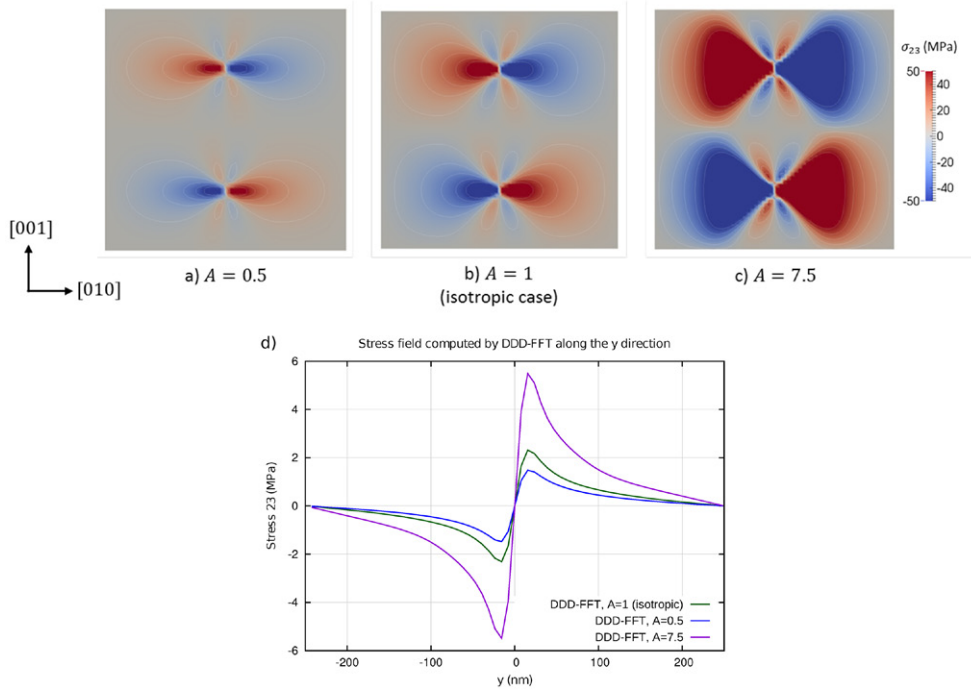


Figure 10. σ_{23} shear component of the stress field of a prismatic loop along a (100) slice taken at $x = a/2$ (see figure 5(a)) for different anisotropic ratios: (a) $A = 0.5$ (b) $A = 1$ (isotropic case) and (c) $A = 7.5$. (d) Evolution of the σ_{23} component normalized with $\mu b/l_0$ as a function of the distance to the dislocation core. The results have been obtained for a Fourier grid made of $64 \times 64 \times 64$ voxels. Note that the computational costs associated with cases (a)–(c) are identical using the DDD-FFT approach.

Several measures on the elastic constants can be associated with the deviation from elastic isotropy. Among them, the anisotropy ratio defined as:

$$A = \frac{2C_{44}}{(C_{11} - C_{12})} \quad (29)$$

is commonly taken as a quantification of the anisotropy of cubic crystals, where $A = 1$ is characteristic for isotropy [27]. For the prismatic loop examined in section 4.1, the isotropic material parameters chosen as $\mu = 51$ GPa and $\nu = 0.37$ correspond to the elastic constants of $C_{11} = 247$ GPa, $C_{12} = 145$ GPa, and $C_{44} = (C_{11} - C_{12})/2 = 51$ GPa. In order to provide insights into the current approach's ability to treat anisotropic elasticity, the effect of anisotropy ratios on the resulting stress of the prismatic loop is compared here. To cover a wide range of possible departures from isotropy, three different anisotropy ratios have been selected: (a) $A = 0.5$ ($C_{44} = 25.5$ GPa) which would correspond to Nb crystals [27], (b) $A = 1$ ($C_{44} = 51$ GPa) corresponding to an isotropic case, and (c) $A = 7.5$ ($C_{44} = 382$ GPa), which is found for α -Fe at high temperatures [32]. The σ_{23} components of the resulting stresses for the configuration depicted in figure 5 are reported in figure 10. In comparing figures 10(a)–(c), the change in the shape of the stress fields induced by the variation of the anisotropy ratio is clearly visible, as attested by the isocontours delineated by the thin white lines, going from an elliptical-like shape for $A = 0.5$ to a triangular-like shape for $A = 7.5$. In figure 10(d), where the stresses are

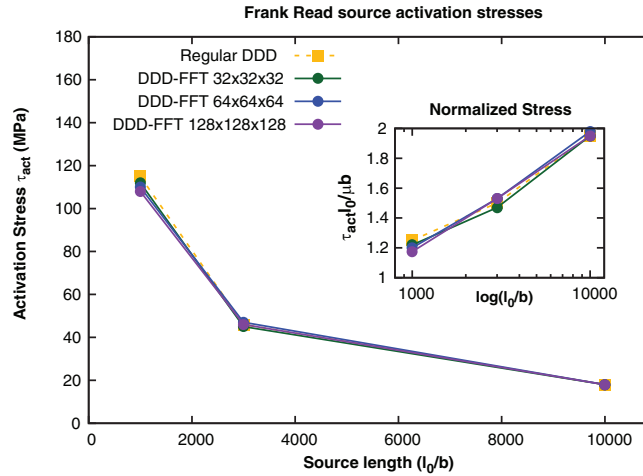


Figure 11. Activation stresses τ_{act} of Frank Read sources of different lengths obtained with the regular DDD approach used in [22] and the DDD-FFT approach for different grid sizes. In the inset, the activation stresses τ_{act} are normalized with $\mu b/l_0$ and reported on a logarithmic scale.

normalized by $\mu b/l_0$, the increase in the stress magnitude associated with the increase in the anisotropy ratio is even more apparent. Note that the computational cost associated with each of these three cases is identical when using the DDD-FFT approach.

4.3. Frank read source activation

To assess the validity of the approach in dynamic cases, the activation stress of a Frank Read source is investigated here. As mentioned earlier, when using the DCM approach, all dislocations present in the simulation volume must be introduced as loops via the Volterra-like process. One simple way to create a realistic initial microstructure in which dislocation sources are present consists in letting the initial loops relax. During this process, the formation of junctions physically pins the dislocations, thereby circumventing the necessity of introducing artificial pinning points as usually employed to initiate dislocation sources. In this section, for the sake of simplicity, the Frank Read source is artificially created by immobilizing three of the four edge segments of the prismatic loop depicted in figure 5(a). With this, the only remaining mobile segment will act as an edge source pinned at its end points. Here, the activation stresses of prismatic loops of size ranging from $l_0 = 1000b$ to $l_0 = 10000b$ are examined. The Burgers vector $\mathbf{b} = [010]$ is chosen such that $|\mathbf{b}| = b = 0.3$ nm and the loops are introduced into the cubic simulation volumes of sides $a = 5l_0$ discretized into $32 \times 32 \times 32$, $64 \times 64 \times 64$ and $128 \times 128 \times 128$ grid points. Only the bottom edge segment is set mobile such that the Frank Read source glides on the (001) plane. The material is assumed to be elastically isotropic with shear modulus $\mu = 130$ GPa and Poisson's ratio $\nu = 0.309$. In this setting, the dislocation source is solely driven by the σ_{23} stress component and produces a ϵ_{23}^p shear. The values obtained for the activation stresses τ_{act} when using the regular DDD code and the DDD-FFT approach are reported in figure 11. This results show that the Frank Read source activates at similar stresses in all cases, thereby assessing the ability of the present method to deal with dynamics cases. Essentially, obtaining similar activation stresses as when using the regular DDD method in which internal stresses are analytically calculated demonstrates the capacity

of the present approach to efficiently capture the evolution of internal stresses directly from the motion of dislocation segments.

4.4. Performance of the DDD-FFT approach

The overall efficiency of the DDD-FFT code primarily results from the use of the very efficient FFT algorithm to compute the mechanical fields. For a DDD simulation volume discretized according to a regular Fourier grid composed of N voxels in each spatial direction, i.e. for $N_{\text{tot}} = N^3$ total number of grid points, the complexity associated with the computation of the discrete Fourier transforms and its inverse (as required in section 3.1) scales with $O(N_{\text{tot}} \log N_{\text{tot}})$ while the computation of the strain field in the Fourier space as defined in equation (15) is a $O(N_{\text{tot}})$ process. Comparatively, a complexity of $O(N_{\text{tot}}^2)$ can be achieved in the most favorable case with the standard 3D FEM (assuming simple linear elements), where N_{tot} is the total number of elements, i.e. for comparable resolutions. Obviously, for large values of N_{tot} , the difference between $N_{\text{tot}} \log N_{\text{tot}}$ and N_{tot}^2 becomes significant, hence the gain in the computation time up to several orders of magnitudes is generally attributed to FFT-based approaches.

In parallel, by coupling the FFT-solver with the DCM procedure, the DDD-FFT approach benefits from the significant reduction in dislocation–dislocation elastic interaction calculations compared to most of the current DDD approaches. Thus, in the case where sufficiently fine grids are used, the DDD complexity is significantly less dependent on the number of dislocation segments N_{seg} present in the volume, but more on the number of grid points N_{tot} , i.e. on the resolution. The numerical complexity associated with the DCM procedure alone is an $O(N_{\text{seg}})$ process requiring considerably fewer numerical operations than computing dislocation–dislocation interactions (even with an analytical formulation), since it solely consists in analytically determining the intersection area between the dislocations sheared area and the elementary spheres of their neighboring grid points, as detailed in section 3.2. In the current implementation, the algorithm to compute the intersection areas (see appendix C) is able to process a million intersections in 0.2 s. Furthermore, as mentioned earlier, the computation of supplementary local contributions requires the determination of pairs of interacting portions of dislocation segments whose distances are closer than $h/2 = L$. The analytical method presented in appendix B to determine such portions of segments is implemented so as to process a million segment pairs in 0.5 s.

Figure 12 shows a comparison of the execution run times per time step as a function of the number of segments N_{seg} and of the DDD approach used for a periodic DDD simulation on an elastically isotropic material. Here, the regular DDD refers to the version of the DDD code used in [22], where the nodal forces on neighboring dislocation segments are analytically computed from the non-singular formulation developed in [26], while the box method technique is employed to compute far-field elastic interactions [51]. The computation of the far-field elastic interactions via the use of the box method is performed every 10 simulation steps. The averaged execution times per simulation step obtained using the regular DDD approach with different numbers of boxes N_{box} are compared to those obtained with the DDD-FFT approach for different grid sizes N_{tot} . For the sake of comparison, the run time corresponding to a full $O(N_{\text{seg}}^2)$ calculation is also reported. For a fair comparison, i.e. not accounting for the complications associated with parallel computation, the execution times have been measured in running the code on a single CPU.

From figure 12 it is clear that the DDD-FFT becomes very efficient when large numbers of segments are to be treated, hence highlighting the potential of the approach. Note also that

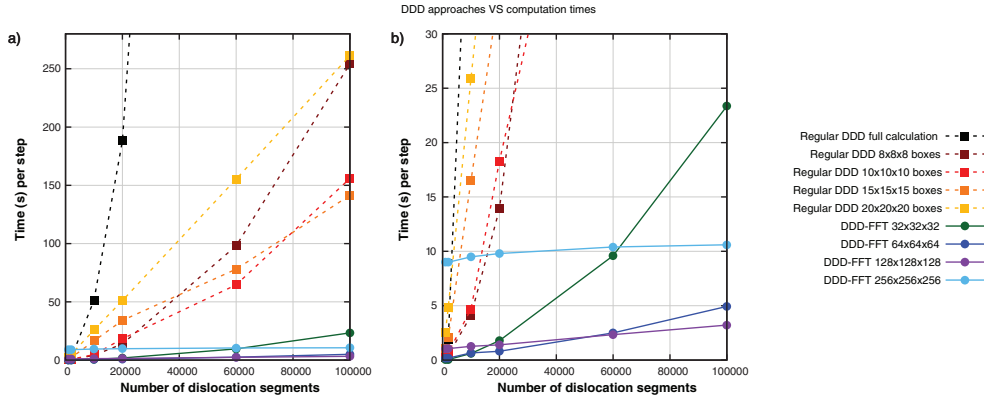


Figure 12. (a) Comparison of the computational cost between regular DDD simulations and the DDD-FFT approach as a function of the number of segments in the simulation volume for elastically isotropic materials. The times, averaged over several simulation steps, are given for the utilization of a single CPU and measured according to the current implementation of the DDD code. The times of the regular DDD simulations are given based on the full calculation and on the utilization of the box method [51] for different numbers of boxes. The times for the DDD-FFT approach are given for different grid sizes. (b) Close-up of the computation times obtained with the DDD-FFT approach for different grid sizes.

in order to obtain the same precision with the regular DDD approach as with the DDD-FFT approach, the box method would have to be performed at every simulation step. In practice, to reduce the computation time, it is generally called at a specified frequency, since the elastic field of far dislocations is not expected to drastically change from one step to another. In the computation times reported in figure 12, the box method is performed every 10 steps. If it were to be performed at every step, i.e. working with the same precision as with the DDD-FFT approach, the computation times of the regular DDD method reported in figure 12 would roughly increase by a factor of 10. A close-up of the DDD-FFT computation times as a function of the grid size is given in figure 12(b). Interestingly, the computation time associated with the grid made of $32 \times 32 \times 32$ voxels becomes significantly more important than that of finer meshes as the number of segments N_{seg} increases. This is because the number of local interactions becomes all the more prevalent than the mesh is coarse when N_{seg} increases. For the same reason, the computation cost for a grid of $64 \times 64 \times 64$ voxels becomes more expensive than that for $128 \times 128 \times 128$ voxels for $N_{\text{seg}} > 50\,000$ in the current implementation. Therefore, for optimal performances, the size of the grid should be chosen as a function of the maximum number of segments to be treated, if such quantity is known *a priori*. A grid refinement technique may be implemented to dynamically select the optimal resolution [59]. For $N_{\text{seg}} = 100\,000$, the computation times for grids of $64 \times 64 \times 64$ and $128 \times 128 \times 128$ voxels remain under 5 s per simulation step, thereby allowing for a gain in time of a factor up to 30 compared to the regular DDD method when using the box method. Finally, note that for a grid of $256 \times 256 \times 256$ voxels, the computation time ranges between 9 and 11 s according to the number of segments. For that grid size, the FFT solver alone (including the computation of both Fourier transforms and that of the strain in the Fourier space) requires about 8 s. While the actual computation of local interactions is insignificant for such a mesh size, the sole search for the portions of segments closer than $h/2$ accounts for the rest of the time.

Note, however, that the more efficient Fast Multipole Method (FMM) has progressively replaced the Box Method approximation in DDD simulations [4, 31, 52, 60]. Since the authors

have not implemented this technique at the present time, a direct comparison is difficult. However, through the implementation of their new version of the FMM, Zhao and co-workers reported a gain of a factor 21 in the isotropic stress calculation when using the FMM compared to the full calculation for $N_{\text{seg}} = 25\,000$ [60]. In the present work, the gain in the total step time compared to the full calculation amounts to 230 when using the DDD-FFT with $64 \times 64 \times 64$ voxels and to 140 for $128 \times 128 \times 128$ voxels for $N_{\text{seg}} = 20\,000$.

Furthermore, the results reported in figure 12 are given for simulations using isotropic elasticity. In the case of anisotropic elasticity, the computational gain is expected to amount to a supplementary order of magnitude, since the overall cost is expected to be comparable to that of isotropic elasticity for fine meshes. In contrast, anisotropic calculations are rarely used in regular DDD simulations due to their prohibitive cost. For instance, in previous works [29, 30, 61], the relative cost of anisotropic calculations was reported to amount to between 200 and 500 times that of isotropic calculations. However, the recent development of approximations based on expansions in spherical harmonics have permitted significant gains in computational time, but their efficiency remains largely conditioned by the amount of anisotropy and the desired level of accuracy, and their relative cost still remains above one order of magnitude [32, 62].

Finally, note that the times reported here are strongly dependent on the current numerical implementation and optimizations of the DDD code, as well as on the software and hardware configurations, and are therefore meant to provide a basis for comparison.

5. Discussion

In section 4, the validity and the efficiency of the DDD-FFT approach for simulating plasticity in the bulk of homogeneous isotropic and anisotropic materials under PBC is demonstrated. Furthermore, in the case where the Fourier grid is chosen to be sufficiently fine such that the computation of local contributions pertaining to neighboring segments seldom occurs, the gain in computation time with respect to the regular DDD approach is significant when a large number of segments are to be treated. As stated earlier, local contributions still have to be added for portions of segment pairs closer than the mesh size $L = h/2$. The mean spacing of dislocations typically scales as $\lambda = 1/\sqrt{\rho}$ where ρ is the dislocation density. Therefore, assuming that all dislocation segments are equally spaced within the volume, the number of local contributions to compute is not expected to be significant up to dislocation densities of $\rho = 1/L^2$. For instance, for a simulation volume of $V_s = (5\,\mu\text{m})^3$ discretized into $64 \times 64 \times 64$ voxels, the DDD-FFT approach is theoretically expected to perform well up to densities of $\rho = 10^{14}\,\text{m}^{-2}$. Thus, simulations of realistic microstructure can be performed on desktop computers without requiring a large number of CPUs. Conversely, when the Fourier grid is coarse, the DDD-FFT approach becomes analogous to a DDD simulation in which elastic far fields are computed via the FMM in which stress fields are inherently computed from infinite replications of the primary volume. Besides, the performances reported in figure 12 are voluntarily given when running the code on a single CPU. However, note that the DDD-FFT approach would be fully and easily parallelizable, especially thanks to the efficient parallel 3D FFT implementations that have been developed. In addition, it is interesting to notice that the calculation of FFT is particularly suitable for GPU-accelerated computing, thereby enabling a further acceleration of the present approach.

Furthermore, the ability of FFT-based approaches to handle elastic heterogeneities offers broad new possibilities to DDD simulations. Practically, as originally done in FFT-based approaches for studying composite materials [41, 42] and polycrystalline aggregates [43, 44],

the primary DDD simulation volume can easily be partitioned into grains with different elastic properties. Note that the FEM-based implementation of the DCM approach proposed in the work of Vattré *et al* [39] already provides such a possibility. However, the application to realistic simulations is not comparable to the DDD-FFT approach in terms of achievable level of strains and accuracy. Therefore, the method presented in this work paves the way towards achieving a practical scale transition between DDD scale plasticity and crystal plasticity. In particular, DDD can be regarded as a tool to test and validate constitutive laws. Furthermore, the present approach establishes a direct connection between DDD simulations and Field Dislocation Mechanics models, for which spectral methods were recently introduced for static cases [56, 58]. In that sense, the DDD-FFT can be regarded as a 3D dynamic extension of these works.

The principal limitation of the present formulation lies in its prerequisite to deal with fully periodic simulations, i.e. with simulations for which PBCs are prescribed in all spatial directions. In other words, the present formulation of the DDD-FFT approach cannot be used to simulate bounded simulations in which free-surfaces are to be accounted for. However, the recent development of spectral methods to account for the effects of free-surfaces [9] may be adapted to the DDD-FFT approach as a way to overcome the limitation associated with the periodicity. Implementing these is beyond the scope of the present work, and shall be undertaken in a separate study.

6. Conclusion

In this paper, a new full-field approach to computing mechanical fields in periodic DDD simulations is presented. The main idea of the DDD-FFT approach relies on the computation of strains and stresses induced by the presence of dislocation segments in a periodic simulation volume directly in the Fourier space, while the transformation between the real Cartesian space and the Fourier space is performed thanks to the FFT algorithm. Furthermore, by coupling the DDD-FFT approach with the DCM approach, the stress state obtained in the Fourier grid benefits from the inherent incorporation of long-range elastic interactions. In addition, it is demonstrated that the approach allows for the treatment of anisotropic elasticity at a relatively similar cost than that of isotropic elasticity. Specifically, the DCM approach requires the computation of supplementary local dislocation–dislocation interactions that are not accounted for by the grid, and whose number scales with the mesh size. Hence, in the case where the simulation volume is discretized into a sufficiently fine mesh such that the number of supplementary local interactions to compute reduces to a minimum, the computational cost would become insensitive to the shape of the elasticity tensor. Moreover, when compared to the current approaches used in DDD for isotropic elasticity, it is shown that the DDD-FFT approach becomes significantly more efficient when large number of segments are to be treated. For instance, the DDD-FFT approach allows for a gain of at least a factor 20 compared to the regular DDD implementation in [22] for 100 000 segments. In the case of anisotropic elasticity, the computational gain is expected to a supplementary order of magnitude, since the overall cost is expected to be comparable to that of isotropic elasticity for fine meshes. Although the recent development of spherical harmonics approximations have permitted significant gains in anisotropic calculations, their efficiency remains largely conditioned by the amount of anisotropy and the desired level of accuracy [32, 62]. Finally, with the ability of FFT-based formulations to account for heterogeneous elasticity, it is to be expected that the method presented here could enable fast DDD simulations on polycrystals.

Acknowledgments

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Appendix A

In the context of spectral methods, the strain solution $\boldsymbol{\epsilon}(\mathbf{x})$ in equation (16) obtained in the real space from taking the inverse Fourier transform of equation (15) gives the fluctuations from the mean strain $\boldsymbol{\epsilon}$, where $\langle \cdot \rangle$ denotes the spatial average over the primary volume V_s . Therefore, in the FFT-based formulation, driving a DDD simulation by imposing a macroscopic strain rate $\dot{\mathbf{E}}$ is achieved by computing the total strain $\boldsymbol{\epsilon}_{t+dt}(\mathbf{x})$ at each time step $t + dt$ from equation (16) as:

$$\begin{aligned}\boldsymbol{\epsilon}_{t+dt}(\mathbf{x}) &= \mathcal{FFT}^{-1}(\hat{\boldsymbol{\epsilon}}_{t+dt}(\boldsymbol{\xi})) + \mathbf{E}_{t+dt} \\ \hat{\boldsymbol{\epsilon}}(\mathbf{0}) &= \mathbf{0}\end{aligned}\tag{A.1}$$

where $\hat{\boldsymbol{\epsilon}}_{t+dt}(\boldsymbol{\xi})$ is the strain field in the Fourier space calculated from the plastic strain $\boldsymbol{\epsilon}_{t+dt}^p(\mathbf{x})$ regularized at step $t + dt$, and $\mathbf{E}_{t+dt} = \mathbf{E}_t + \dot{\mathbf{E}}dt$ is the imposed total strain. Equation (A.1) is valid under the condition $\hat{\boldsymbol{\epsilon}}(\mathbf{0}) = \mathbf{0}$ imposed on the average strain in equation (15). As a result, the stress $\boldsymbol{\sigma}_{t+dt}(\mathbf{x})$ at step $t + dt$ is computed from Hooke's law as:

$$\boldsymbol{\sigma}_{t+dt}(\mathbf{x}) = \mathbb{C}(\boldsymbol{\epsilon}_{t+dt}(\mathbf{x}) - \boldsymbol{\epsilon}_{t+dt}^p(\mathbf{x}))\tag{A.2}$$

and encompasses both the contributions of the applied loading and the internal stress fields induced by the presence of dislocation lines within the simulation volume V_s and its replica. By definition, for any arbitrary dislocation configuration, the stress $\boldsymbol{\sigma}(\mathbf{x})$ obtained with equation (A.2) for $\mathbf{E} = \mathbf{0}$ (superscript $t + dt$ is dropped in the following for the sake of simplicity), i.e. in the case where no loading is imposed, corresponds to the internal stresses produced by the dislocation segments in an infinite free medium. Therefore, let us denote $\boldsymbol{\sigma}^{\text{int}}(\mathbf{x}) = \boldsymbol{\sigma}(\mathbf{x})$ the stress obtained in (A.2) by the DDD-FFT method under no applied loading. An important characteristic of the present FFT-based formulation is that the internal stress field $\boldsymbol{\sigma}^{\text{int}}$ —outside of dislocation cores—is not expected to coincide with the analytical solution $\tilde{\boldsymbol{\sigma}}^{\text{int}}$ that would be obtained for the same load-free microstructure. Here the analytical solution $\tilde{\boldsymbol{\sigma}}^{\text{int}}$ refers to the stress field obtained by the superposition of the stresses induced by individual dislocation segments and their images using the line integral (7) formula or the non-singular approach developed by Cai *et al* [26] when dealing with isotropic elasticity (the non-singular formulation is based on the Mura formula and matches it outside of the dislocation cores). This difference finds its origin in the fact that the imposed loading does not have the same physical meaning in both approaches: in the FFT-based formulation, the loading is imposed by fixing the value of the average strain $\mathbf{E} = \langle \boldsymbol{\epsilon} \rangle$ (or stress) corresponding to a macroscopic loading. Thus, a load-free microstructure in the DDD-FFT approach corresponds to a microstructure for which $\mathbf{E} = \langle \boldsymbol{\epsilon} \rangle = \mathbf{0}$ is imposed, which is not the case when using the analytical formulation. In other words, the boundary value problem is not solved for the same boundary conditions in both approaches. As a result, since the only difference between both approaches lies in the value of the average total strain, the stress field $\boldsymbol{\sigma}^{\text{int}}$ obtained with the FFT-based approach for an arbitrary microstructure differs from the analytical solution $\tilde{\boldsymbol{\sigma}}^{\text{int}}$ by a constant translation tensor $\bar{\boldsymbol{\sigma}}$ such that $\boldsymbol{\sigma}^{\text{int}}(\mathbf{x}) = \tilde{\boldsymbol{\sigma}}^{\text{int}}(\mathbf{x}) + \bar{\boldsymbol{\sigma}}, \forall \mathbf{x} \in V_s$.

Note, however, that the analytical and the FFT-based solutions would coincide for a microstructure yielding $\langle \boldsymbol{\epsilon}^p \rangle = \mathbf{0}$. The condition $\langle \boldsymbol{\epsilon}^p \rangle = \mathbf{0}$ physically manifests by the presence of a dipolar configuration: in that case, $\tilde{\boldsymbol{\sigma}}^{\text{int}}$ reduces to $\mathbf{0}$ in the analytical solution, and as a result

$\langle \tilde{\epsilon} \rangle \equiv \mathbb{C}^{-1} \langle \tilde{\sigma}^{\text{int}} \rangle + \langle \epsilon^p \rangle = \mathbf{0}$ would match the FFT-based solution for $\langle \epsilon \rangle = \mathbf{E} = \mathbf{0}$. With that in mind, it follows that $\sigma^{\text{int}}(\mathbf{x})$ and $\tilde{\sigma}^{\text{int}}(\mathbf{x})$ are not expected to coincide during a DDD-FFT simulation, since the evolution of an arbitrary microstructure does not yield $\langle \epsilon^p \rangle = 0$ in general.

In addition, as mentioned in section 2.3, a supplementary local stress contribution must be computed for segments whose interaction distance is smaller than half of the regularization parameter h , since the grid does not account for such close field interactions. The analytical solution for the stress field of dislocation segments can be used in order to compute such supplementary local contributions. In the case where the computation of the local contributions is required, for the sake of consistency, both the stress obtained in equation (17) and the analytical stress must coincide, i.e. $\tilde{\sigma}$ must be determined. Since the translation tensor $\tilde{\sigma}$ is constant and hence does not depend on the spatial position on the grid, it can be easily calculated by comparing $\sigma^{\text{int}}(\mathbf{x})$ and the analytical stress solution $\tilde{\sigma}^{\text{int}}(\mathbf{x})$ at any arbitrary point $\mathbf{x}$ in the simulation volume. Therefore, in the current formulation, the determination of $\tilde{\sigma}$ requires the computation of the analytical stress solution at one arbitrary point in the volume at each simulation step. More details on the computation of supplementary local forces are given in appendix B.

Appendix B

In a nodal DDD code, the total force $\mathbf{F}_i$ acting at dislocation node i is assembled as [4]:

$$\mathbf{F}_i = \sum_j \mathbf{f}_{ij} \quad (\text{B.1})$$

where $\mathbf{f}_{ij}$ denotes the force on dislocation segment ij acting at node i . The summation in equation (B.1) is performed over all nodes j connected to node i . For each dislocation segment ij defined between end nodes $\mathbf{x}_i$ and $\mathbf{x}_j$ and with burgers vector $\mathbf{b}_{ij}$, the force $\mathbf{f}_{ij}$ is obtained by integration of the Peach–Koehler force $\mathbf{f}_{ij}^{pk}(\mathbf{x})$ along its length as:

$$\mathbf{f}_{ij} = \int_{\mathbf{x}_i}^{\mathbf{x}_j} N_i(\mathbf{x}) \mathbf{f}_{ij}^{pk}(\mathbf{x}) d\mathbf{x} \quad (\text{B.2})$$

where $N_i(\mathbf{x})$ denotes the interpolation function associated with node i and where the Peach–Koehler force along segment ij given in equation (3) is recalled here:

$$\mathbf{f}_{ij}^{pk}(\mathbf{x}) = [\boldsymbol{\sigma}(\mathbf{x}) \cdot \mathbf{b}_{ij}] \times \mathbf{t}_{ij} \quad (\text{B.3})$$

where $\mathbf{t}_{ij} = (\mathbf{x}_j - \mathbf{x}_i)/l_{ij}$ is the unit tangent of segment ij of length $l_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$. In equation (B.3), the total stress $\boldsymbol{\sigma}(\mathbf{x})$ at point $\mathbf{x}$ must encompass all contributions from internal and external stresses. In the DDD-FFT approach, the stress obtained in equation (17) already includes both contributions thanks to the coupling with the DCM. However, because the grid does not account for local interactions between portions of dislocation lines closer than $h/2$, where h is the regularization parameter, local interactions must be added as a supplementary contribution. Therefore, equation (B.2) can be expressed as:

$$\mathbf{f}_{ij} = \int_{\mathbf{x}_i}^{\mathbf{x}_j} N_i(\mathbf{x}) [(\boldsymbol{\sigma}(\mathbf{x}) \cdot \mathbf{b}_{ij}) \times \mathbf{t}_{ij}] d\mathbf{x} + \mathbf{f}_{ij}^{\text{loc}} \quad (\text{B.4})$$

where $\boldsymbol{\sigma}(\mathbf{x})$ is the stress in equation (17) computed via the FFT-based approach and $\mathbf{f}_{ij}^{\text{loc}}$ denotes the local force arising from the supplementary local interactions. By considering all

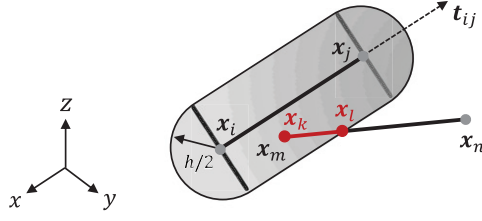


Figure B1. Schematic of portions of neighboring segments that must be accounted for as supplementary local interactions. The red sub-segment kl corresponds to the portion of neighboring segment mn whose distance to segment ij is closer than $h/2$. End points x_k and x_l can be analytically determined by computing the intersections between segment mn and (1) the finite cylinder of radius $h/2$ and of axis t_{ij} bounded by normal planes at x_i and x_j , and (2) the two half spheres of radius $h/2$ centered in x_i and x_j .

portions kl of neighboring dislocation lines whose distance to dislocation segment ij is smaller than $h/2$, the local force can be expressed as:

$$f_{ij}^{\text{loc}} = \sum_{kl} f_{ij}^{kl} \quad (\text{B.5})$$

where f_{ij}^{kl} denotes the interaction force between segment ij and sub-segment kl defined between end nodes x_k and x_l . As depicted in figure B1, sub-segments kl correspond to portions of neighboring dislocation segments mn included in the union of spheres of radius $h/2$ centered in x when x spans dislocation segment ij , i.e. for $x = (1-s)x_i + sx_j$, $s \in [0, 1]$. In order to avoid any computational burden associated with numerical integration, the determination of sub-segments kl can be analytically performed by computing the intersections between (1) the finite cylinder of axis t_{ij} and bounded at $s = 0$ and $s = 1$, and (2) the two half spheres of radius $h/2$ centered in x_i and x_j , respectively. Note that, as described in section 3.2, the regularization parameter is set such that $h/2 = L$ in the current approach.

With that, the local force f_{ij}^{kl} can be computed by the integration of the analytical stress $\tilde{\sigma}^{kl}$ induced by segment kl along segment ij . As in appendix A, the analytical stress here refers to the stress field induced by a dislocation and its replica in an infinite medium, as given by the Mura formula (7) or by the non-singular formulation [26] for isotropic elasticity. However, as mentioned in section 3.1 and detailed in appendix A, for any segment kl , the stress $\sigma^{kl}(x)$ computed from the FFT-based approach is obtained up to a constant translation tensor $\bar{\sigma}$ with respect to the analytical solution $\tilde{\sigma}^{kl}(x)$ such that $\sigma^{kl}(x) = \tilde{\sigma}^{kl}(x) + \bar{\sigma}$, $\forall x \in V_s$. As a result, a translation force must to be accounted for, such that the local force can be decomposed as:

$$f_{ij}^{kl} = \tilde{f}_{ij}^{kl} + \bar{f}_{ij}^{kl} \quad (\text{B.6})$$

where $\tilde{f}_{ij}^{kl}$, the nodal force for the pair of segment ij defined between end points x_i and x_j and sub-segment kl defined between end points x_k and x_l , is given by:

$$\tilde{f}_{ij}^{kl} = \int_{x_i}^{x_j} N_i(x) [(\tilde{\sigma}^{kl}(x) \cdot b_{ij}) \times t_{ij}] dx \quad (\text{B.7})$$

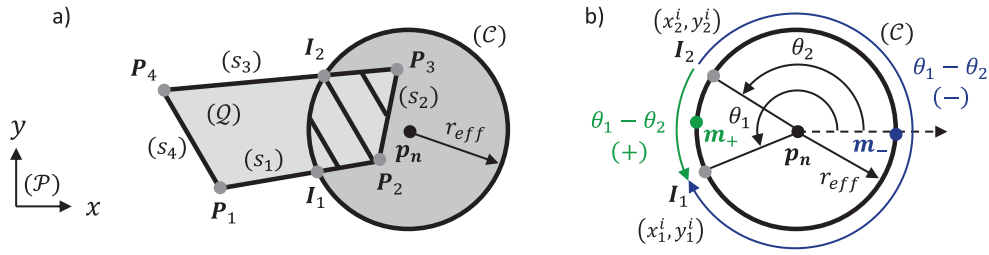


Figure C1. Schematic of the intersection between quadrilateral Q defined by the sheared area produced by the glide of a dislocation segment on a plane $\mathcal{P}$ and circle $\mathcal{C}$ of radius r_{eff} resulting from the intersection between the sphere $d\phi(\mathbf{p})$ and plane $\mathcal{P}$. (a) The intersection area $dS_{ij}^p = \text{area}(Q \cap \mathcal{C})$ (shaded region) can be analytically calculated using Green's theorem by following the oriented contour composed of the straight segments $\overline{I_1P_2}$, $\overline{P_2P_3}$ and $\overline{P_3I_2}$ and the arc $\widehat{I_2I_1}$, where $\{P_j\}_{j=1,4}$ denotes the vertices of Q and $\{I_k\}_{k=1,n}$ the n intersection between edges of Q and $\mathcal{C}$ (here $n = 2$). (b) Arc $\widehat{I_2I_1}$ can possibly be travelled along the positive (+) or the negative (−) direction. When $n = 2$, the direction along which it must be travelled can be determined as that whose middle point, respectively m_+ and m_- , lies within quadrilateral Q . In the present case, $m_+ \in Q$ and the arc should be travelled in the positive (+) direction.

for which an analytical expression is given in [4] for isotropic elasticity, and $\tilde{\mathbf{f}}_{ij}^{kl}$ is the translation force given by:

$$\tilde{\mathbf{f}}_{ij}^{kl} = \int_{x_i}^{x_j} N_i(\mathbf{x}) [(\tilde{\boldsymbol{\sigma}} \cdot \mathbf{b}_{ij}) \times \mathbf{t}_{ij}] d\mathbf{x} = \frac{1}{2} l_{ij} [(\tilde{\boldsymbol{\sigma}} \cdot \mathbf{b}_{ij}) \times \mathbf{t}_{ij}] \quad (\text{B.8})$$

When dealing with anisotropic elasticity, nodal force $\tilde{\mathbf{f}}_{ij}^{kl}$ in equation (B.7) can be obtained using recent efficient methods [30, 32] or by numerical integration of the anisotropic stress fields given in equation (7) [28, 29].

Appendix C

As introduced in section 3.2, the regularization of the plastic strain involved in the DCM procedure requires the computation of the intersection areas dS_{ij}^p between the sheared area produced by the glide of each individual dislocation segment ij and the elementary spheres $d\phi(\mathbf{p})$ of radius $h/2$ defined around each grid point $\mathbf{p}$, according to equation (19) (see figure 2(a)).

Let us consider a dislocation segment ij defined between end nodes i and j moving from their initial positions $\mathbf{x}_i$ and $\mathbf{x}_j$ at velocities $\mathbf{v}_i$ and $\mathbf{v}_j$ during time dt , respectively, and gliding on plane $\mathcal{P}$ with unit normal $\mathbf{n}$. First, the sheared area produced by the glide of the segment forms a quadrilateral Q defined by vertices $\{P_1, P_2, P_3, P_4\} = \{\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_j + \mathbf{v}_j dt, \mathbf{x}_i + \mathbf{v}_i dt\}$ (see figure 2(a)). Second, the intersection between the sphere $d\phi(\mathbf{p})$ and the glide plane $\mathcal{P}$ results in a circle $\mathcal{C}$ of radius $r_{\text{eff}} = \sqrt{h^2/4 - d^2}$ centered at $\mathbf{p}_n$, where $\mathbf{p}_n$ is the orthogonal projection of $\mathbf{p}$ onto plane $\mathcal{P}$ such that $\mathbf{p} - \mathbf{p}_n = d\mathbf{n}$. Therefore, as depicted in figure C1, the intersection area dS_{ij}^p (shaded region) is defined by the area of the region $Q \cap \mathcal{C}$.

The value of dS_{ij}^p can be analytically determined using Green's theorem (equation (21)) once the contour of $Q \cap \mathcal{C}$ is determined. This contour can be solely composed of straight

segments and arcs. Numerically, its determination can be achieved using the method described hereafter. First note that $(\mathcal{Q} \cap \mathcal{C}) \in \mathcal{P}$ such that the problem can be conveniently solved in 2D space. Let us select a frame $(\mathcal{O}, x, y)$ on plane $\mathcal{P}$ in which the coordinates of vertices $\{\mathbf{P}_j\}_{j=1,4}$ are given by (x_j^p, y_j^p) . The first step to determine the contour of $\mathcal{Q} \cap \mathcal{C}$ consists in sequentially finding the intersections points $\{\mathbf{I}_k\}_{k=1,n}$ of coordinates (x_k^i, y_k^i) between the oriented edges $\{s_j\}_{j=1,4}$ of $\mathcal{Q}$ and the circle $\mathcal{C}$, where s_j denotes the segment defined between points $\mathbf{P}_j$ and $\mathbf{P}_{j+1}$. Since each edge s_j may not intersect with circle $\mathcal{C}$, intersect in one point, or intersect in two points, the total number of intersections n can take values of $0 \leq n \leq 8$. Region $\mathcal{Q} \cap \mathcal{C}$ is then delimited by joining the intersection points $\{\mathbf{I}_k\}_{k=1,n}$ and the quadrilateral vertices $\{\mathbf{P}_j\}_{j=1,4}$ through straight segments and arcs. With that, area dS_{ij}^p is computed using expression (21), in which straight segments and arcs line integrals are calculated using equations (22) and (23), respectively.

The only remaining difficulty lies in the determination of the angle $\theta_k - \theta_j$ (see equation (23)) in the case of arc portions $\widehat{\mathbf{I}_j \mathbf{I}_k}$, as a circle can possibly be travelled along two different paths from coordinates (x_j^i, y_j^i) to (x_k^i, y_k^i) , namely along the positive (+) or negative (−) direction. This difficulty can easily be tackled in the case where more than two intersections are detected (i.e. for $n > 2$): in that case, the arc defined between two intersection points should be travelled along the path that does not include any other intersection point $\mathbf{I}_l$. If the only detected intersection points are those defining the arc, the middle point of each possible path can be tested to choose the path for which the middle point lies inside the quadrilateral $\mathcal{Q}$ (see figure C1(b)). The latter condition can be checked using a point-in-polygon (PIP) algorithm based on the ray casting approach.

Note that in the case where no intersection between the circle and the quadrilateral are found (i.e. for $n = 0$), either the circle is entirely comprised into the quadrilateral, or the quadrilateral is fully comprised into the circle, or both are well separated. Such configurations can be easily identified, especially via the use of bounding boxes and PIP techniques.

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